

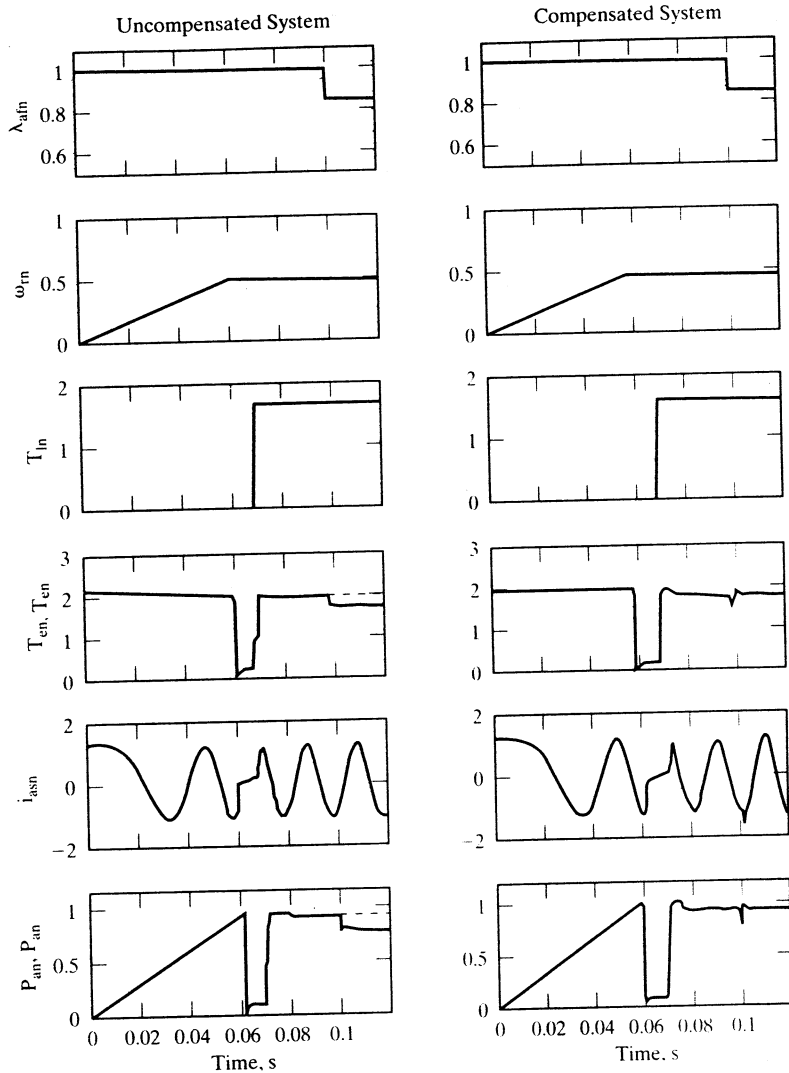


Figure 9.46 Simulation results of the speed-controlled drive system with constant torque angle for a step change in λ_{af}

Speed-Controlled-Drive Performance: The second set of simulations is for a speed-controlled drive system with a step change in its speed from 0 to 0.5 p.u. applied initially; then, a step change of load torque from 0 to 1.65 p.u. is applied when the speed equals its command. Finally, λ_{af} is changed from nominal to 0.85 times the nominal value. The results for both the uncompensated and compensated speed-controlled drive systems are shown in Figure 9.46. The reference torque T_e^* is limited to 2 p.u. In the uncompensated system, the change in rotor flux linkages

causes a reduction in torque and hence in the speed, momentarily. This increases the torque reference through the action of the outer speed-feedback loop, resulting in an increase in the air gap power reference (as is shown by dotted lines). The q axis stator current increases to meet the load torque resulting in the speed's being maintained equal to its reference in steady state. In the compensated system, the torque reference and air gap power reference do not change, but the stator q axis current is increased until it delivers torque equal to its reference. Resulting compensation has the advantage in that it does not change torque reference and hence will not lead to a speed reduction, as it can in the uncompensated system. For example, consider that the torque reference is altered by the speed feedback loop in the uncompensated system from 2 to 2.5 p.u., with the torque limiter having a maximum of 2 p.u. That means the stator q axis current will be generated only for 2 p.u. torque reference, hence generating less than 2 p.u. torque with the reduction in rotor flux linkages. If load torque is 2 p.u., then the speed will have to come down. That this is not the case in the compensated system is obvious.

Flux-Weakening-Drive Performance: The third set of simulations is for the speed-controlled drive system operating in the flux-weakening region. The condition for the simulation is to have a step change of λ_{af} from nominal to 0.85 times the nominal value with the drive system operating in the flux-weakening mode. The simulation results are shown in Figure 9.47. The flux-weakening process is initiated in this drive when the speed is greater than 0.5 p.u., where the reference torque T_e^* and hence the actual torque start falling as expected until the commanded speed is reached. When a step change is applied to λ_{af} , a rise in reference torque is observed, but the actual torque does not rise to match the reference torque, because of a drop in rotor flux linkages. The compensated system, however, shows a perfect matching of the torque and its reference, via almost instantaneous stator current compensation.

Effect of Quadrature-Axis Inductance Variation: The final set of simulations shows the step change of L_q from nominal to 0.85 times the nominal value in Figure 9.48, including the operation in the flux-weakening mode. The variation of L_q would not have any effect on a constant-torque-angle-controlled system where $\delta = 90^\circ$, because the controller parameters and the output torque are not dependent on the quadrature-axis inductance, but it does affect the system in the flux-weakening mode: both the controller parameters and the torque output are dependent on the quadrature-axis inductance. The compensated system, however, corrects for changes in L_q , as is shown in Figure 9.48. The effects are similar to the case where λ_{af} changes.

Effect of Stator-Resistance Variation: Since the air gap power calculator is dependent on the stator resistance, which changes with temperature, simulations for a step change of stator resistance are necessary to verify the adequacy of the air gap-power-compensation scheme to neutralize the variations in stator resistance. The effects were seen to be negligible on the compensated system. The stator resistance

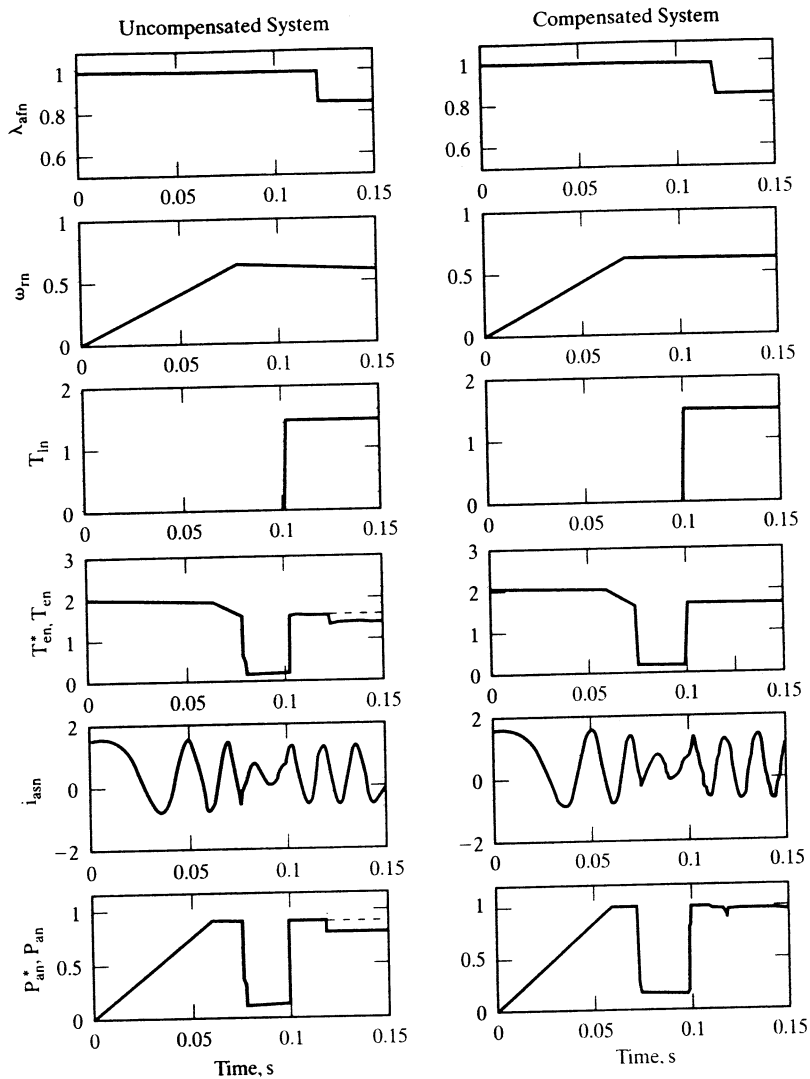


Figure 9.47 Simulation results of a speed-controlled drive system operating in the flux-weakening mode with a step change in λ_{af}

variations can also be compensated for via direct temperature measurement. The quadrature-axis inductance variation can be compensated with the q axis current magnitude. No direct monitoring of the rotor temperature or rotor flux linkages is possible to compensate for rotor-flux-linkages variation. Hence, schemes such as the air gap-power feedback control are necessary to overcome the parameter sensitivity of the rotor-flux-linkages variation, and the same then could be used to compensate for other motor-parameter variations.

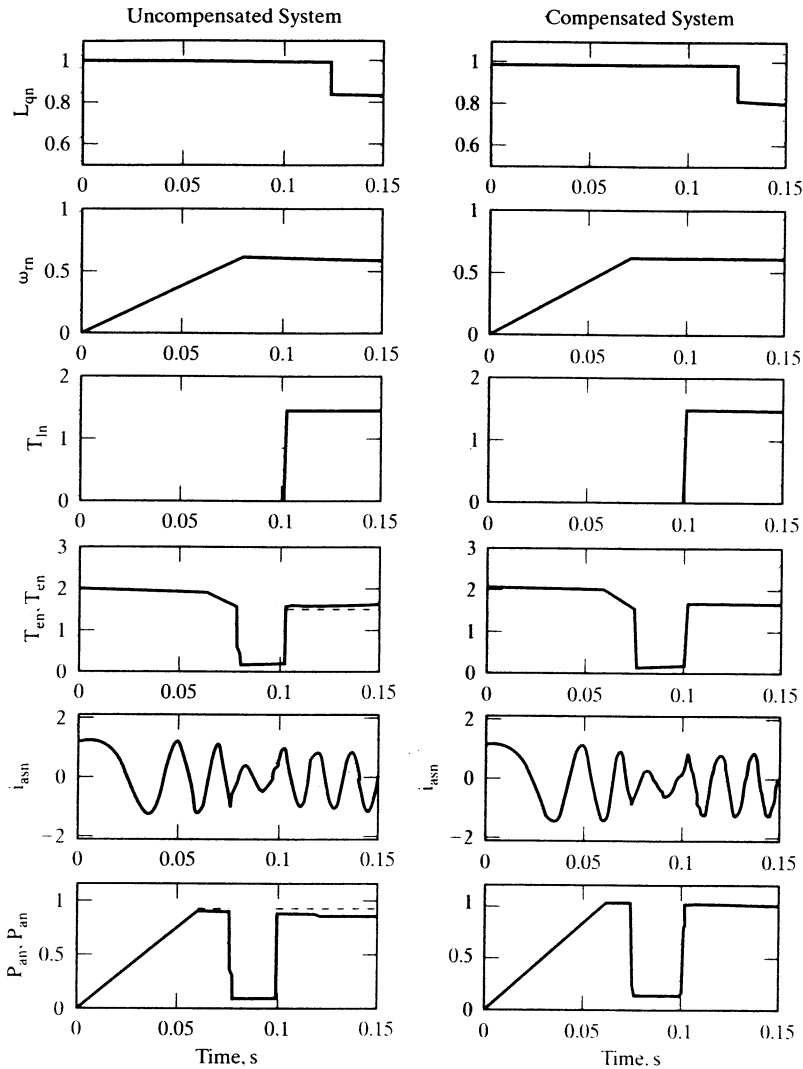


Figure 9.48 Simulation results of the effect of variation of L_q on the speed-controlled drive system, including the operation in the flux-weakening mode

9.10 PM BRUSHLESS DC MOTOR (PMBDCM)

PM synchronous machines having trapezoidal induced emf are known as PM brushless dc machines. The advantage of such a machine in comparison to the PMSM was discussed in the early part of this chapter. The major reason for the popularity of these machines over their counterparts is control simplicity. To initiate the onset and commutation of current in the phase of a machine, the beginning and end of the constant portion of the induced emf have to be tracked. That amounts to only six discrete positions for a three-phase machine in each electrical cycle. These signals could easily be

generated with three Hall sensors displaced from each other by 120 electrical degrees. The Hall sensors are mounted facing a small magnet wheel fixed to the rotor and having the same number of poles as the rotor of the PMBDCM, or the extra magnet wheel may be dispensed with by extending the rotor beyond the stack length of the stator and using the rotor magnets to provide the position information. Such an arrangement tracks the absolute position of the rotor magnets and hence the shape and position of the induced emfs in all the machine phases. In contrast to the PMSM, which requires continuous and instantaneous absolute rotor position, the PMBDCM position-feedback requirement is much simpler: it requires only six discrete absolute positions for a three-phase machine, resulting in a major cost saving in the feedback sensor. Further, the control involves significant vector operations in the PMSM drive, whereas such operations are not required for operation of the PMBDCM drive.

9.10.1 Modeling of PM Brushless DC Motor

The flux distribution in a PM brushless dc motor is trapezoidal; therefore, the d - q rotor reference frames model developed for the PM synchronous motor is not applicable. Given the nonsinusoidal flux distribution, it is prudent to derive a model of the PMBDCM in phase variables. The derivation of this model is based on the assumptions that the induced currents in the rotor due to stator harmonic fields are neglected and that iron and stray losses are also neglected. Damper windings are not usually a part of the PMBDCM; damping is provided by the inverter control. The motor is considered to have three phases, even though the derivation procedure is valid for any number of phases.

The coupled circuit equations of the stator windings in terms of motor electrical constants are

$$\begin{bmatrix} v_{as} \\ v_{bs} \\ v_{cs} \end{bmatrix} = \begin{bmatrix} R_s & 0 & 0 \\ 0 & R_s & 0 \\ 0 & 0 & R_s \end{bmatrix} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} + p \begin{bmatrix} L_{aa} & L_{ab} & L_{ac} \\ L_{ba} & L_{bb} & L_{bc} \\ L_{ca} & L_{cb} & L_{cc} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_{as} \\ e_{bs} \\ e_{cs} \end{bmatrix} \quad (9.180)$$

where R_s is the stator resistance per phase, assumed to be equal for all three phases. The induced emfs e_{as} , e_{bs} , and e_{cs} are all assumed to be trapezoidal, as shown in Figure 9.8, where E_p is the peak value, derived as

$$E_p = (Blv)N = N(Blr\omega_m) = N\phi_a\omega_m = \lambda_p\omega_m \quad (9.181)$$

where N is the number of conductors in series per phase, v is the velocity, l is the length of the conductor, r is the radius of the rotor bore, ω_m is the angular velocity, and B is the flux density of the field in which the conductors are placed. This flux density is solely due to the rotor magnets. The product (Blr) , denoted as ϕ_a , has the dimensions of flux and is directly proportional to the air gap flux, ϕ_g :

$$\phi_a = Blr = \frac{1}{\pi} B\pi l r = \frac{1}{\pi} \phi_g \quad (9.182)$$

Note that the product of flux and number of conductors in series has the dimension of flux linkages and is denoted by λ_p . Since this is proportional to phase a flux linkages by a factor of $\frac{1}{\pi}$, it is hereafter referred to as modified flux linkages.

If there is no change in the rotor reluctance with angle because of a nonsalient rotor, and assuming three symmetric phases, the following are obtained:

$$L_{aa} = L_{bb} = L_{cc} = L; \text{ and } L_{ab} = L_{ba} = L_{ac} = L_{ca} = L_{bc} = L_{cb} = M(H) \quad (9.183)$$

Substituting equations (9.182) and (9.183) in equation (9.180) gives the PMBDCM model as

$$\begin{bmatrix} v_{as} \\ v_{bs} \\ v_{cs} \end{bmatrix} = R_s \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} + \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix} p \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_{as} \\ e_{bs} \\ e_{cs} \end{bmatrix} \quad (9.184)$$

The stator phase currents are constrained to be balanced, i.e., $i_{as} + i_{bs} + i_{cs} = 0$, which leads to the simplification of the inductance matrix in the model as

$$\begin{bmatrix} v_{as} \\ v_{bs} \\ v_{cs} \end{bmatrix} = \begin{bmatrix} R_s & 0 & 0 \\ 0 & R_s & 0 \\ 0 & 0 & R_s \end{bmatrix} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} + \begin{bmatrix} (L - M) & 0 & 0 \\ 0 & (L - M) & 0 \\ 0 & 0 & (L - M) \end{bmatrix} p \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_{as} \\ e_{bs} \\ e_{cs} \end{bmatrix} \quad (9.185)$$

The electromagnetic torque is given by

$$T_e = [e_{as}i_{as} + e_{bs}i_{bs} + e_{cs}i_{cs}] \frac{1}{\omega_m} \text{ (N}\cdot\text{m)} \quad (9.186)$$

The instantaneous induced emfs can be written from Figure 9.8 and equation (9.181) as

$$e_{as} = f_{as}(\theta_r) \lambda_p \omega_m \quad (9.187)$$

$$e_{bs} = f_{bs}(\theta_r) \lambda_p \omega_m \quad (9.188)$$

$$e_{cs} = f_{cs}(\theta_r) \lambda_p \omega_m \quad (9.189)$$

where the functions $f_{as}(\theta_r)$, $f_{bs}(\theta_r)$, and $f_{cs}(\theta_r)$ have the same shape as e_{as} , e_{bs} , and e_{cs} , with a maximum magnitude of ± 1 . The induced emfs do not have sharp corners, as is shown in trapezoidal functions, but rounded edges. The emfs are the result of the flux-linkages derivatives, and the flux linkages are continuous functions. Fringing also makes the flux density functions smooth with no abrupt edges. The electromagnetic torque then is

$$T_e = \lambda_p [f_{as}(\theta_r)i_{as} + f_{bs}(\theta_r)i_{bs} + f_{cs}(\theta_r)i_{cs}] \text{ (N}\cdot\text{m)} \quad (9.190)$$

It is significant to observe that the phase-voltage equation is identical to the armature-voltage equation of a dc machine. That is one of the reasons for naming this machine the *PM brushless* dc machine. The equation of motion for a simple system with inertia J , friction coefficient B , and load torque T_l is

$$J \frac{d\omega_m}{dt} + B\omega_m = (T_e - T_l) \quad (9.191)$$

and electrical rotor speed and position are related by

$$\frac{d\theta_r}{dt} = \frac{P}{2} \omega_m \quad (9.192)$$

Combining all the relevant equations, the system in state-space form is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (9.193)$$

where

$$\mathbf{x} = [i_{as} \quad i_{bs} \quad i_{cs} \quad \omega_m \quad \theta_r]^T \quad (9.194)$$

$$\mathbf{A} = \begin{bmatrix} -\frac{R_s}{L_1} & 0 & 0 & -\frac{\lambda_p}{L_1} f_{as}(\theta_r) & 0 \\ 0 & -\frac{R_s}{L_1} & 0 & -\frac{\lambda_p}{L_1} f_{bs}(\theta_r) & 0 \\ 0 & 0 & -\frac{R_s}{L_1} & -\frac{\lambda_p}{L_1} f_{cs}(\theta_r) & 0 \\ \frac{\lambda_p}{J} f_{as}(\theta_r) & \frac{\lambda_p}{J} f_{bs}(\theta_r) & \frac{\lambda_p}{J} f_{cs}(\theta_r) & -B/J & 0 \\ 0 & 0 & 0 & \frac{P}{2} & 0 \end{bmatrix} \quad (9.195)$$

$$\mathbf{B} = \begin{bmatrix} \frac{1}{L_1} & 0 & 0 & 0 \\ 0 & \frac{1}{L_1} & 0 & 0 \\ 0 & 0 & \frac{1}{L_1} & 0 \\ 0 & 0 & 0 & -\frac{1}{J} \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (9.196)$$

$$L_1 = L - M \quad (9.197)$$

$$\mathbf{u} = [v_{as} \quad v_{bs} \quad v_{cs} \quad T_l]^T \quad (9.198)$$

The state variable θ_r , rotor position, is required so as to have the functions $f_{as}(\theta_r)$, $f_{bs}(\theta_r)$, and $f_{cs}(\theta_r)$ which can be realized from a stored table. This completes the modeling of the PMBDCM.

9.10.2 The PMBDCM Drive Scheme

For constant-torque operation at speeds lower than the base speed, this drive requires six discrete pieces of position information. They correspond to every 60 electrical degrees for energizing the three stator phases, as shown in Figure 9.8.

The flux-weakening is slightly different for this motor and is discussed later. The control scheme for the PMBDM drive is simple and is shown in Figure 9.49. The resolver gives absolute rotor position; it is converted into rotor speed through the signal processor. The rotor speed is compared to its reference, and the rotor speed error is amplified through the speed controller. The output of the speed controller

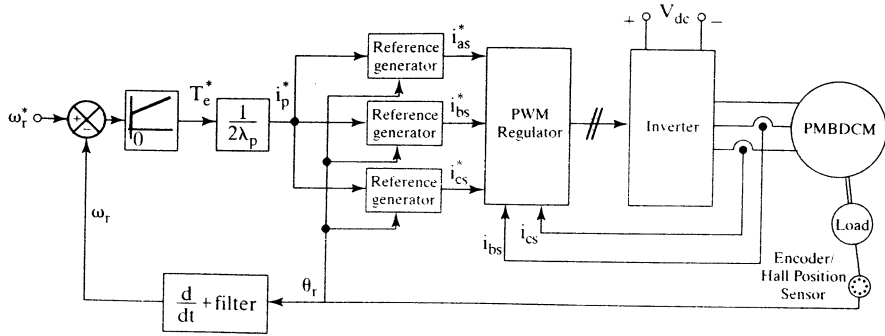


Figure 9.49 Speed-controlled PMBDM drive scheme without flux-weakening

provides the reference torque, T_e^* . The current-magnitude command, i_p^* , is obtained from the torque expression as

$$T_e^* = \lambda_p [f_{as}(\theta_r)i_{as}^* + f_{bs}(\theta_r)i_{bs}^* + f_{cs}(\theta_r)i_{cs}^*] \quad (9.199)$$

Only two machine phases conduct current at any time, with the two phases being in series for full-wave inverter operation, so the phase currents are equal in magnitude but opposite in sign. The rotor-position-dependent functions have the same signs as the stator phase currents in the motoring mode, but opposite signs in the regeneration mode. The result of such sign relationships is simplification of torque command as,

$$T_e^* = 2\lambda_p i_p^* \quad (9.200)$$

The stator-current command is derived from (9.200) as

$$i_p^* = \frac{T_e^*}{2\lambda_p} \quad (9.201)$$

The individual stator-phase current commands are generated from the current-magnitude command and absolute rotor position. These current commands are amplified through the inverter by comparing them with their respective currents in the stator phases. Only two phase currents are necessary in the balanced three-phase system to obtain the third phase current, since the sum of the three phase currents is zero.

The current errors are amplified and used with pulse-width modulation or hysteresis logic to produce the switching-logic signals for the inverter switches, as explained in Chapter 4.

9.10.3 Dynamic Simulation

The simulation results for a step speed reference input of from 0 to 1 p.u. are shown in Figure 9.50 for the PWM current controllers. The rotor is at standstill at time zero. With the onset of speed reference, the speed error and torque references attain a maximum value, which is limited to 2.0 p.u in this case. The current is made to follow the reference by the current controller. Therefore, the electromagnetic torque follows its

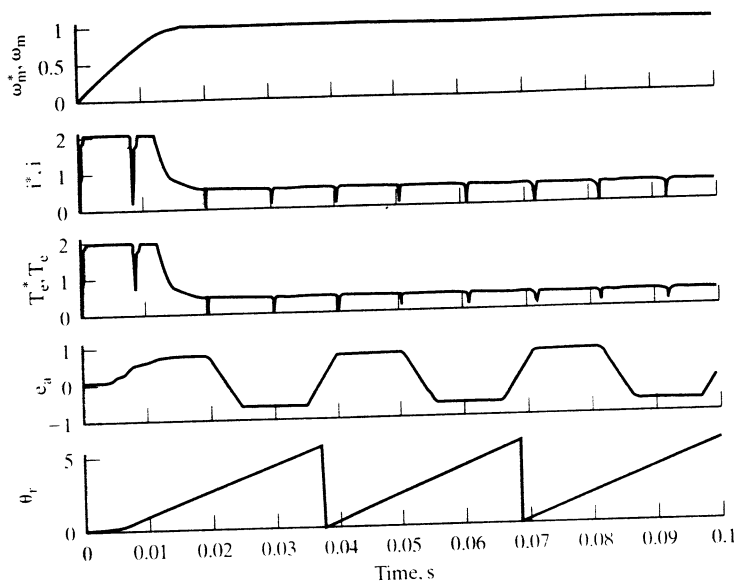


Figure 9.50 Performance of a speed-controlled PMBDCM drive system with PWM current controllers

reference very closely. The ripples on the torque are due to current ripples produced by the switching. The PWM control considered in this case is operating at 2 kHz, and the current controller has an amplification of 100.

9.10.4 Commutation-Torque Ripple

The desired current waveform is rectangular and 120° wide in each half-cycle for a three-phase PMBDCM drive. The leakage inductance, L_l , causes the stator currents to take a finite time to rise and fall, thus distorting the ideal waveform into a trapezoidal shape. The effect of this is the torque ripple generated at the current transitions. For a three-phase machine, there will be six torque ripples for every 360° electrical, as the six current transitions occur. They will also reduce the average torque if the conduction time is maintained at 120° electrical, whereby the constant-current region is reduced below 120° electrical. The consequences of a set of practical currents on the performance of the PMBDCM drive can be analyzed by using a Fourier-series approach as follows. The torque expressions are given considering only a two-pole machine; for a P-pole machine, the expressions have to be multiplied by the number of pole pairs.

The phase current is generalized, as shown in Figure 9.51. The phase a current can be resolved into Fourier series as

$$i_{as}(\theta_r) = \frac{4I_p}{\pi(\theta_2 - \theta_1)} \left[(\sin \theta_2 - \sin \theta_1) \sin \theta_r + \frac{1}{3^2} (\sin 3\theta_2 - \sin 3\theta_1) \sin 3\theta_r + \dots \right] \quad (9.202)$$

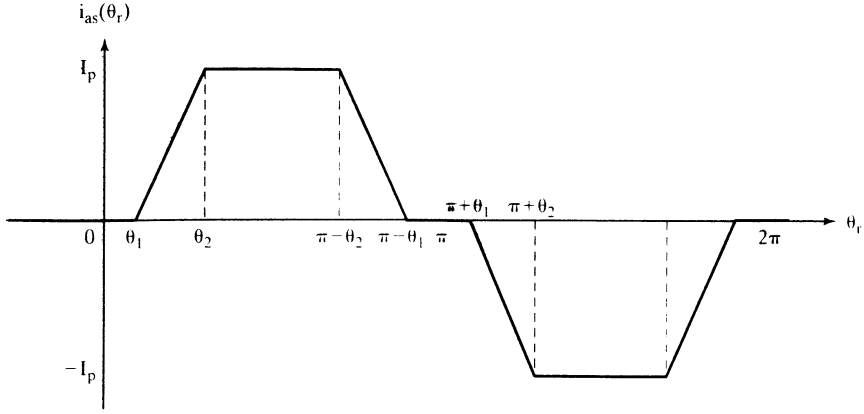


Figure 9.51 General phase-current waveform of the PMBDCM

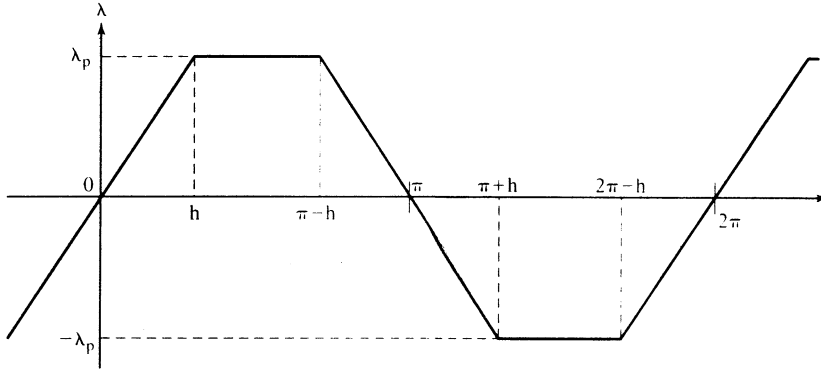


Figure 9.52 The rotor-flux-linkage waveform

Similarly, the Fourier series of the flux linkages of phase a , assuming a trapezoidal waveform and constancy for $(\pi - 2h)$ electrical degrees in each half-cycle, is

$$\lambda_{at}(\theta_r) = \frac{4\lambda_p}{\pi h} \left[\sin h \sin \theta_r + \frac{1}{3^2} (\sin 3h \sin 3\theta_r) + \frac{1}{5^2} (\sin 5h \sin 5\theta_r) + \dots \right] \quad (9.203)$$

where λ_p is the peak value of the modified flux linkages and the flux linkage waveform is shown in Figure 9.52.

Similarly, the b and c phase currents and their modified flux linkages can be derived.

The fundamental electromagnetic torque is computed by considering the product of fundamental terms in the air gap flux linkages and respective stator currents for a 2-pole machine:

$$T_{cl} = \lambda_{at1}(\theta_r) i_{as1}(\theta_r) + \lambda_{bt1}(\theta_r) i_{bs1}(\theta_r) + \lambda_{ct1}(\theta_r) i_{cs1}(\theta_r) \quad (\text{N} \cdot \text{m}) \quad (9.204)$$

Substituting the fundamental terms and expanding the expressions yields

$$T_{e1} = \frac{16I_p\lambda_p}{\pi^2h(\theta_2 - \theta_1)} [\sin h(\sin \theta_2 - \sin \theta_1)\sin^2 \theta_r + \sin h(\sin \theta_2 - \sin \theta_1)\sin^2(\theta_r - 2\pi/3) + \sin h(\sin \theta_2 - \sin \theta_1)\sin^2(\theta_r + 2\pi/3)] \quad (9.205)$$

For $h = \pi/6$, the electromagnetic torque for three phases is evaluated as

$$T_{e1} = [I_p\lambda_p] \left\{ \frac{48}{\pi^3(\theta_2 - \theta_1)} (\sin \theta_2 - \sin \theta_1) \right\} \left[\frac{3}{2} \right] = 2.3193 \left[\frac{\sin \theta_2 - \sin \theta_1}{\theta_2 - \theta_1} \right] [I_p\lambda_p] \text{ (N}\cdot\text{m)} \quad (9.206)$$

The normalized fundamental torque in p.u. as a function of θ_2 for θ_1 and h equal to 30° is shown in Figure 9.53, for a 1-p.u current. This shows that increasing the rise time of the current decreases the fundamental torque. At higher speeds, for the same rise time of the current, note that θ_2 increases; hence, there will be a greater reduction in the fundamental torque of the motor drive.

For 120° electrical rectangular current, the fundamental torque is

$$T_{e1} = 2.011I_p\lambda_p \quad (9.207)$$

which closely approximates the available torque calculated from Figure 9.8.

The commutation torque is at six times the fundamental frequency. It can be seen as the result of the sum of the fundamental rotor flux linkages interacting with the fifth- and seventh-current harmonics and the fundamental of the current interacting with the fifth- and seventh-harmonic rotor flux linkages. It is derived as follows.

$$T_{e6} = \frac{4\lambda_p I_p}{\pi(\theta_2 - \theta_1)} \left[(\sin \theta_2 - \sin \theta_1) \left\{ -\frac{1}{5^2} (\sin 5h) + \frac{1}{7^2} (\sin 7h) \right\} \right] \frac{4}{\pi h} + \frac{4}{\pi h} \cdot \frac{4}{\pi(\theta_2 - \theta_1)} \left[\sin h \left\{ -\frac{1}{5^2} (\sin 5\theta_2 - \sin 5\theta_1) + \frac{1}{7^2} (\sin 7\theta_2 - \sin 7\theta_1) \right\} \right]$$

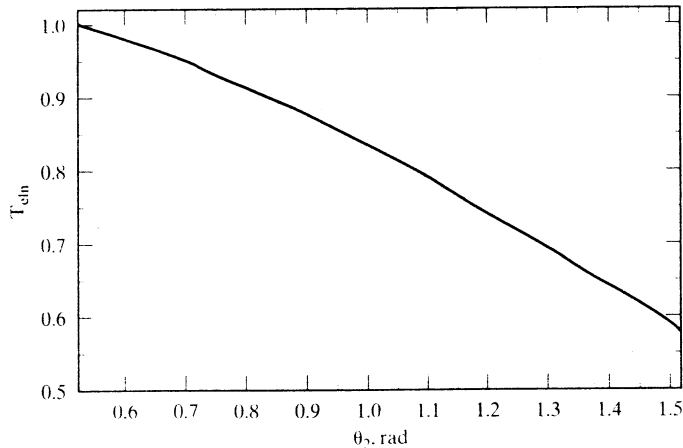


Figure 9.53 The fundamental torque vs. θ_2 for $\theta_1 = 0.524$ (rad) and $h = 0.524$ (rad) for $I_p = 1$ p.u.

$$= \frac{16\lambda_p I_p}{\pi^2 h (\theta_2 - \theta_1)} \left[(\sin \theta_2 - \sin \theta_1) \left\{ -\frac{1}{5^2} \sin 5h + \frac{1}{7^2} \sin 7h \right\} \right. \\ \left. + \sin h \left\{ -\frac{1}{5^2} (\sin 5\theta_2 - \sin 5\theta_1) + \frac{1}{7^2} (\sin 7\theta_2 - \sin 7\theta_1) \right\} \right] \quad (9.208)$$

In general, the harmonic torque of frequency m times the fundamental is given by

$$T_{em} = \frac{16\lambda_p I_p}{\pi^2 h (\theta_2 - \theta_1)} \left[(\sin \theta_2 - \sin \theta_1) \left\{ -\frac{1}{(m-1)^2} \sin \overline{m-1}h + \frac{1}{(m+1)^2} \sin \overline{m+1}h \right\} \right. \\ \left. + \sin h \left\{ -\frac{1}{(m-1)^2} (\sin \overline{m-1}\theta_2 - \sin \overline{m-1}\theta_1) + \frac{1}{(m+1)^2} (\sin \overline{m+1}\theta_2 - \sin \overline{m+1}\theta_1) \right\} \right] \\ m = 6, 12, 18, 24, \dots \quad (9.209)$$

Note that this is for one phase only. The negative sign attributed to the fifth-harmonic rotor flux linkage and current is due to the fact that they are revolving against the fundamental rotor flux linkage and current; hence, their torque contributions are considered negative. The detailed calculation for the sixth- and twelfth-harmonic torques are given in Tables 9.1 and 9.2, respectively.

The harmonic torques are normalized on the basis of the fundamental torque for a perfect rectangular current. It is rational from the point of view of

TABLE 9.1 Sixth-harmonic torque for various θ_2 with $\theta_1 = 30^\circ = h$

Harmonic Number		Sixth-Harmonic Torque Component					
Rotor Flux Linkages	Stator Current	$\theta_2 = 32.5^\circ$	35°	37.5°	40°	45°	50°
1	5	0.424	0.439	0.447	0.448	0.428	0.383
1	7	-0.258	-0.221	-0.178	-0.132	-0.037	0.044
5	1	-0.079	-0.078	-0.077	-0.076	-0.075	-0.071
7	1	-0.041	-0.040	-0.039	-0.039	-0.037	-0.036
Total T_{e6n}		0.047	0.101	0.153	0.202	0.280	0.320

TABLE 9.2 Twelfth-harmonic torque for $\theta_1 = 30^\circ = h$

Harmonic Number		Sixth-Harmonic Torque Component					
Rotor Flux Linkages	Stator Current	$\theta_2 = 32.5^\circ$	35°	37.5°	40°	45°	50°
1	11	-0.201	-0.203	-0.190	-0.163	-0.088	-0.018
1	13	0.122	0.078	0.031	-0.012	-0.063	-0.057
11	1	0.016	0.016	0.016	0.016	0.015	0.015
13	1	0.012	0.012	0.011	0.011	0.011	0.010
Total T_{e12n}		-0.051	-0.097	-0.131	-0.149	-0.126	-0.05

application considerations. Hence, the normalized harmonic torques for the three phases are

$$T_{emh} = \frac{3T_{em}}{T_{el}} = \frac{24}{\pi^2 h(\theta_2 - \theta_1)} \left[(\sin \theta_2 - \sin \theta_1) \left\{ -\frac{1}{(m-1)^2} \sin \overline{m-1h} + \frac{1}{(m+1)^2} \sin \overline{m+1h} \right\} \right. \\ \left. + \sin h \left\{ -\frac{1}{(m-1)^2} (\sin \overline{m-1\theta_2} - \sin \overline{m-1\theta_1}) + \frac{1}{(m+1)^2} (\sin \overline{m+1\theta_2} - \sin \overline{m+1\theta_1}) \right\} \right] \\ \text{for } m = 6, 12, 18, \dots \quad (9.210)$$

9.10.5 Phase Advancing

The magnitude of the induced emfs increases with increasing speed. When the magnitude of the line-to-line induced emf comes close to the magnitude of the dc-link voltage, it will not be possible to maintain the currents at desired level. Then, the currents can be advanced in phase so that the line-to-line induced emf is lower than the dc-link voltage, facilitating the establishment of the desired current magnitude. As the line-to-line induced emf becomes greater than the dc-link voltage, the current cannot flow from the dc link to the motor windings. The energy stored in the leakage inductance, (L-M), will keep the current circulating in the windings through the free-wheeling diodes of the inverter. During this interval, the currents will decrease. This translates into a reduction of torque. If the speed is increased beyond rated value, then the power output can be maintained at rated value for a small range even with decreasing torque production. Note that this mode of operation is usually attempted for a very small speed range. The phase-advancing effect is quantified as follows.

Considering an ideal current with 120° constancy and an ideal flux linkage waveform, they can be resolved into harmonic components and written as

$$i_{as}(\theta_r) = \frac{4\sqrt{3}I_p}{2\pi} \left[\sin \theta_r + \frac{1}{5} \sin 5\theta_r + \dots \right] \quad (9.211)$$

$$\lambda_{af1}(\theta_r) = \frac{24\lambda_p}{\pi^2} \left[\frac{1}{2} \sin \theta_r + \frac{1}{9} \sin 3\theta_r + \frac{1}{2} \cdot \frac{1}{25} \sin 5\theta_r + \dots \right] \quad (9.212)$$

Advancing the current by an angle θ_a can be represented as

$$i_{as}(\theta_r + \theta_a) = \frac{4\sqrt{3}}{2\pi} I_p \left[\sin(\theta_r + \theta_a) + \frac{1}{5} \sin \{5(\theta_a + \theta_r)\} + \dots \right] \quad (9.213)$$

Similarly, expressions for the other phase currents, $i_{bs}(\theta_r + \theta_a)$ and $i_{cs}(\theta_r + \theta_a)$, can be written. Substituting these into the torque expression gives the fundamental torque:

$$T_{el} = \lambda_{af1}(\theta_r) i_{as1}(\theta_r + \theta_a) + \lambda_{bf1}(\theta_r) i_{bs1}(\theta_r + \theta_a) + \lambda_{cf1}(\theta_r) i_{cs1}(\theta_r + \theta_a) \\ = \frac{96\sqrt{3}}{4\pi^3} \lambda_p I_p \left[\sin \theta_r \cdot \sin(\theta_r + \theta_a) + \sin(\theta_r - 2\pi/3) \cdot \sin(\theta_r - 2\pi/3 + \theta_a) \right. \\ \left. + \sin(\theta_r + 2\pi/3) \cdot \sin(\theta_r + 2\pi/3 + \theta_a) \right] \\ = 2.0085 \lambda_p I_p \cos \theta_a \quad (9.214)$$

For speeds less than rated value, the advance angle is zero, giving the fundamental torque:

$$\therefore T_{e1} = 2.0085\lambda_p I_p = T_{er} \quad (9.215)$$

where T_{er} is the rated value of the electromagnetic torque.

For speeds higher than rated value, the phase-advance angle, θ_a , is nonzero. Then, the torque can be represented in terms of the rated value and advance angle from equations (9.213) and (9.214) as

$$T_{e1} = T_{er} \cos \theta_a \quad (9.216)$$

Assuming that the base flux linkage is equal to λ_p and the base current is I_b , then the torque in normalized unit is

$$T_{en1} = I_{pn} \cos \theta_a \quad (9.217)$$

where I_{pn} is the normalized peak current in a machine phase.

It is to be noted that these expressions are valid only for ideal current waveforms; in a practical situation, they will deviate from the ideal considerably. The phase-advancing of currents not only decreases the available torque but also drastically increases the harmonic torque. As the current is advanced in one phase, initially the air gap flux linkage is not a constant but a ramp. The result of the interaction is a ramp-shaped torque, giving way to a significant pulsating-torque component. Their quantification can be achieved by following a procedure similar to the one given in the commutation-torque-ripple section.

9.10.6 Normalized System Equations

The equations of the PMBDCM can be normalized by using base voltage V_b , base current I_b , base flux linkages λ_b , base power P_b , and base frequency ω_b . Considering only phase a for normalization, it is achieved as follows.

$$v_{asn} = \frac{v_{as}}{V_b} = \frac{1}{V_b} [Ri_{as} + (L - M)pi_{as} + \lambda_p \omega_m f_{as}(\theta_r)] \quad (9.218)$$

but the base voltage can be written as

$$V_b = I_b \cdot Z_b = \lambda_b \cdot \omega_b \quad (9.219)$$

By substituting this in the voltage equation, the phase a normalized voltage expression is obtained as

$$v_{asn} = R_{an}i_{asn} + (X_{Ln} - X_{Mn}) \frac{p}{\omega_b} i_{asn} + \lambda_n \omega_{mn} f_{as}(\theta_r) \quad (9.220)$$

where

$$R_{an} = \frac{R}{Z_b} \text{ (p.u.)} \quad (9.221)$$

$$X_{Ln} = \frac{\omega_b L}{Z_b} \text{ (p.u.)} \quad (9.222)$$

$$X_{Mn} = \frac{\omega_b M}{Z_b} \text{ (p.u.)} \quad (9.223)$$

$$\lambda_n = \frac{\lambda_p}{\lambda_b} \text{ (p.u.)} \quad (9.224)$$

$$\omega_{mn} = \frac{\omega_m}{\omega_b} \text{ (p.u.)} \quad (9.225)$$

$$T_{eb} = 2\lambda_b I_b \text{ (Nm)} \quad (9.226)$$

$$Z_b = \frac{V_b}{I_b} \text{ (}\Omega\text{)} \quad (9.227)$$

Similarly, the other two phase equations can be derived. The electromechanical equation is derived as

$$T_e = T_l + B\omega_m + J\frac{d\omega_m}{dt} \quad (9.228)$$

and, in normalized form, the electromagnetic torque is given as

$$T_{en} = \frac{T_e}{T_b} = T_{ln} + B_n\omega_{mn} + 2H\omega_{mn} \quad (9.229)$$

where

$$T_{en} = \frac{T_e}{T_b} \text{ (p.u.)} \quad (9.230)$$

$$T_{ln} = \frac{T_l}{T_b} \text{ (p.u.)} \quad (9.231)$$

$$B_n = \frac{B\omega_b^2}{P_b} \text{ (p.u.)} \quad (9.232)$$

$$H = \frac{1}{2} \frac{J\omega_b^2}{P_b(P/2)^2} \text{ (s)} \quad (9.233)$$

The electromagnetic torque in terms of the flux linkages and motor currents is derived as

$$T_e = \frac{(e_{as}i_{as} + e_{bs}i_{bs} + e_{cs}i_{cs})}{\omega_m} = \lambda_p[i_{as}f_{as}(\theta_r) + i_{bs}f_{bs}(\theta_r) + i_{cs}f_{cs}(\theta_r)] \quad (9.234)$$

and in normalized units it is given as

$$T_{en} = \frac{P}{4} [i_{asn}f_{as}(\theta_r) + i_{bsn}f_{bs}(\theta_r) + i_{csn}f_{cs}(\theta_r)] \text{ (p.u.)} \quad (9.235)$$

$$T_b = 2\lambda_p I_p$$

9.10.7 Half-Wave PMBDCM Drives

Several converter topologies emerge if the PMBDCM is operated in the half-wave mode, i.e., each phase is operated for 120 electrical degrees instead of an alternating

current injection for 120 electrical degrees in each half-cycle in a three-phase machine, which is known as the full-wave operational mode. This full-wave operation has been described earlier in previous subsections. This mode of operation provides very limited opportunities for innovation in the inverter topology; six-switch full-bridge topology has stayed on as the most nearly optimal topology thus far. Cost minimization of the PMBDCM drives is of immense interest to the industry at present, with the opening up of a large number of applications to variable-speed operation. Such applications are to be found in HVAC, fans, pumps, washers, dryers, treadmills and other exercise equipment, wheel chairs, people carriers in airport lobbies, golf carts, freezers, refrigerators, automotive, handtools, and small-process drives with velocity control for packaging, bottling, and food-process applications. With the high-volume nature of these applications, cost minimization is of paramount importance, not only to save materials and labor (and possibly, by parts reduction, to enhance the reliability of the product) but also for the fact that without such a cost minimization many of these applications with variable-speed drives can not be realized, almost certainly in the present and probably in the future.

The cost of the motor and controller with single chips has been optimized; the only other subsystem available for optimization is the power converter. Half-wave operation is a major asset in this aspect, because many power-converter topologies are possible with minimum number of switches. Such a reduction in number of power devices has an impact on the reduction in logic power supplies, heat-sink volume, packaging size, enclosure size, and hence the overall cost of the drive system. Three power topologies for half-wave operation of the PMBDCM drive are presented and discussed in this section.

9.10.7.1 Split-supply-converter topology. The power-converter topology with a single switch per phase and minimum number of diodes in the rectifying section, ideal for fractional-horsepower (fhp) PMBDC motor drives, is shown in Figure 9.54. This topology is similar to the split-voltage, single-switch-per-phase topology for the switched-reluctance motor drive, except that in switched-reluctance motor drives

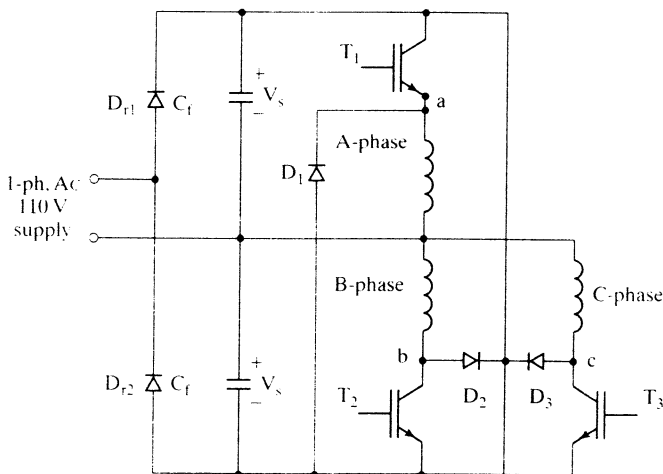


Figure 9.54 Split-supply converter topology

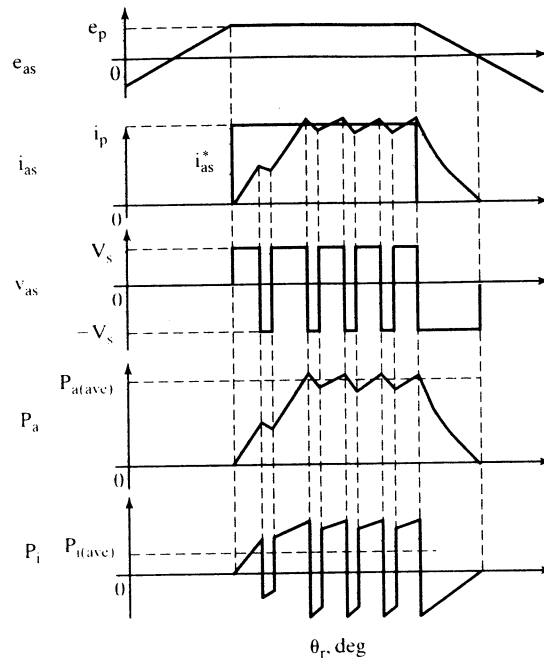


Figure 9.55 First-quadrant operation

it is advocated usually for machines with an even number of phases. The unidirectional current-handling capability of the converter is ideal for switched-reluctance motor drives, but a restriction in the form of half-wave operation is imposed in the PMBDC motor drive with this converter. While such a restriction results in the underutilization of the motor, it leads to other advantages and very welcome features.

By using bifilar windings for the machine, the emitters of the power switches can be tied together and connected to the negative of the dc-link voltage. This simplifies the gate circuitry and further can do away with the split supply (and hence one power capacitor in the dc link), but the bifilar windings would have leakage inductances due to the nonideal coupling of the bifilar windings, resulting in high switching turn-off voltages. The bifilar windings, in addition, also take up slot volume, thus decreasing the area for the main winding, resulting in lower power and torque densities. Before a discussion of merits and demerits of the split-supply converter topology is considered, it is instructive to see the operational modes of this converter with the PMBDCM drive.

A. Operation of the PMBDC Motor with the Split-Supply Converter

Consider the four-quadrant operation of the PMBDCM with the half-wave converter with a restriction of unidirectional current capability. The operation would be taken on a quadrant-by-quadrant basis.

The first quadrant operation is shown in Figure 9.55, where the phase sequence is maintained *abc* with the current injected into the windings when their respective induced emfs experience flat region and for 120° electrical.

Assuming that phase A operation is followed, for example, switch T_1 is turned on at the instant when the stator A-phase induced emf is 30° from its positive-zero

crossover point. Turn-on of the switch T_1 results in the application of voltage V_s to phase winding A , assuming that the switches are ideal. If the phase current exceeds the reference current for that phase, the switch T_1 is turned off, depending on the chosen switching strategy, such as PWM or hysteresis.

During the turn-off of T_1 , the phase current is routed through the diode D_1 , A -phase winding, and the bottom capacitor in the dc link, resulting in a voltage of $-V_s$ applied to the phase A winding. The negative voltage application to the phase reduces the current swiftly by transferring energy from machine phase to the dc link. Thus, the current is controlled between desired limits in the machine phase with only one switch operation. When phase A has to be commutated, note that the switch T_1 is permanently turned off. The average power flow from the dc-link source to the machine winding is positive, indicating that the machine is in motoring mode, i.e., first quadrant. Air gap power P_a and instantaneous input power P_i are shown in Figure 9.55, indicating the same.

To brake when the motor drive is in the forward direction of motion, the electromagnetic torque has to be reversed from positive to negative polarity; thereby, the drive system will be in the fourth quadrant. Since the current cannot be reversed in this converter, the only alternative to get into this mode is to delay the onset of current in machine phases until their induced emfs have negative polarity and 30° from their negative-zero crossover point, as shown in Figure 9.56. This results in

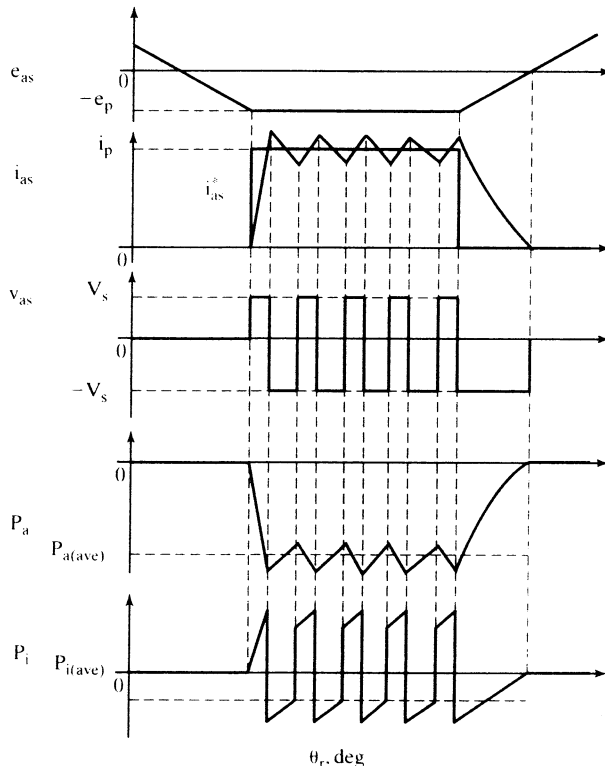


Figure 9.56 Fourth-quadrant operation

TABLE 9.3 Modes of the proposed converter

Mode	T_1	T_2	D_1	D_2	i_{as}	i_{bs}	V_{as}	V_{bs}
1	on	off	off	off	>0	0	V_s	0
2	off	off	on	off	>0	0	$-V_s$	0
3	on	on	on	off	>0	>0	$-V_s$	V_s
4	off	on	on	on	>0	>0	$-V_s$	$-V_s$

negative torque and air gap power in the machine, with the consequence that the drive system is in the fourth quadrant. The current control is very similar to the technique described in the first quadrant.

The third-quadrant operation is reverse motoring; it is obtained by changing the phase sequence to *acb* to obtain the reversal of rotation direction. Other than that, this mode is very similar to the first quadrant. Similar also is the second-quadrant operation to the fourth quadrant, but in the reverse rotational direction. For the sake of brevity, these two quadrants of operation are not considered in detail, even though they are dealt with in the modeling, simulation, and analysis later.

B. Operational Modes of the Converter

The operational modes of the converter are obtained from the above discussion and summarized in Table 9.3. Current conduction in one phase, which usually is the case during most of the operational time, and conduction in two phases, which occurs during the commutation of one phase and energization of the incoming phase, are considered here.

The distinct difference between the full-wave inverter modes and the proposed converter modes is that the proposed converter applies either a positive or a negative voltage across the phase winding when there is a current in it. This has the drawback of higher energy circulation in machine phases, resulting in a small reduction in efficiency and creating torque ripples higher than those in the full-wave-inverter-based PMBDC system that could result in higher acoustic noise in this drive system.

C. Merits and Demerits of the PMBDC Drive with the Split-Supply Converter

The PMBDC motor drive with this converter topology has the following advantages:

- (i) One switch and one diode per phase halves the power-stage requirement of the switches and diodes compared to that of the full-wave inverter topology, resulting in low cost and high compactness in packaging.
- (ii) Reduced gating driver circuits and logic power supplies leads to cost reduction and high compactness in packaging.
- (iii) Full four-quadrant operational capability provides the attendant possibility for high-performance applications, such as in small-process drives. Note that the regenerative brake is to be added, which is common even for the full-wave inverter topology.
- (iv) High reliability is due to the switch being always in series with the machine phase winding, thus preventing a shoot-through fault.

- (v) It is possible to operate with one switch or phase-winding failure, whereas it is not possible under these conditions in the PMBDC drive with full-wave inverter. This is an important and an attractive feature for applications such as wheelchair drives and some process drives.
- (vi) Lower conduction losses are due to one switch and diode per phase as compared to two switches and two diodes being in operation for the full-wave-inverter-fed PMBDC motor drive, resulting in half the losses compared to the full-wave drive, with the current being the same in both the drives. Further, the turn-on losses are reduced by the higher machine inductance in the PMBDC motor for operation with the proposed converter topology, because this allows for a soft turn-on of the power devices.
- (vii) This topology allows for sensing of the induced emf across the machine phases by looking at the switch voltages during their turn-off intervals, and mechanical or optical sensorless operation becomes possible by using these signals to generate the control signals. Note that two-phase switches have a common return, thus doing away with isolation requirement for the transducer signal, and also the fact that two phase voltages are sufficient for generation of the control signals for a three-phase PMBDC motor drive.

The disadvantages of this are as follows:

- (i) Poorer utilization of the motor is due to the half-wave operation. The torque density in terms of the torque per unit stator copper losses is lower in the half-wave-controlled machine by nearly 30% compared to the full-wave-controlled machine.
- (ii) Extra power capacitor is required in the dc link, due to split supply.
- (iii) Larger self-inductance for this PMBDC motor results in large electrical time constant, leading to a slow response in current and hence in the torque compared to the full-wave-inverter-fed PMBDC motor drive.
- (iv) The commutation-torque ripple frequency is halved in that drive compared to that of the full-wave-inverter drive. The commutation torque ripple can be attenuated, as in the full-wave-inverter drive, by coordinating the current in the rising phase with the outgoing phase winding current to yield a constant torque during the phase commutation.

Because of these disadvantages, this topology might not be useful and appropriate for integral-hp drive systems.

D. Design Considerations for the PMBDC Motor

This section contains the design considerations of the PMBDC motor required for use with all the half-wave converter topologies. Wherever possible, all these important factors are contrasted with those of the H-bridge inverter (full-wave)-operated PMBDC motor. For the sake of comparison, the full-wave operated motor is considered as the base.

The following relationships are made on the basis of equal copper volume in the slot of the PMBDC motor. The subscripts l and b correspond to the motor with this converter and the motor with the full-wave converter, respectively.

The ratio of the induced emfs is

$$\frac{e_1}{e_b} = \left(\frac{k_1}{k_b} \right) \left(\frac{\omega_1}{\omega_b} \right) \quad (9.236)$$

where k is the emf constant and ω is the rotor electrical speed.

The air gap power ratio is given by

$$\frac{P_1}{P_b} = \frac{1}{2} \left(\frac{k_1}{k_b} \right) \left(\frac{\omega_1}{\omega_b} \right) \left(\frac{I_1}{I_b} \right) \quad (9.237)$$

where I is the current in the winding.

The ratio of emf constants in terms of the number of turns per phase and conductor cross sections is

$$\frac{k_1}{k_b} = \frac{N_1}{N_b} = \frac{a_b}{a_1} \quad (9.238)$$

where N denotes the number of turns/phase and a is the area of cross section of the conductor in general.

The ratio of the copper losses is given as

$$\frac{P_{cl}}{P_{cb}} = \frac{1}{2} \left[\frac{I_1 N_1}{I_b N_b} \right]^2 = 2 \left[\frac{T_{cl}}{T_{cb}} \right]^2 \quad (9.239)$$

where T_c corresponds to electromagnetic torque in general.

The ratio of the stator resistances is

$$\frac{R_1}{R_b} = \frac{a_b N_1}{a_1 N_b} = \left[\frac{N_1}{N_b} \right]^2 \quad (9.240)$$

From these relationships, it is possible to find the number of turns per phase, electromagnetic torque, induced emf, air gap power, and stator copper losses in the PMBDC motor for the proposed converter topology, depending on the choice of such criteria as equality of torque, equality of copper losses, and equality of air gap power.

A comparison of PMBDC machines for use with half-wave and full-wave inverters is made here. The basis of the comparison is restricted to equal stator phase currents and equal dc-link voltage. Then, three distinct options emerge as follows:

- (i) unequal copper losses, equal volume of copper and, hence, copper fill, and equal maximum speeds;
- (ii) equal copper losses, equal volume of copper and, hence, copper fill, and unequal maximum speeds; and
- (iii) equal copper losses, unequal volume of copper (and, hence, unequal copper fill factors), but such an option increases volume of the copper and possibly stator lamination size, resulting in a large machine size. This option also needs to be considered in applications where the size is not very critical and reliability is of high concern.

From Table 9.4, it is seen that the option (i) has the certain drawback of having twice the copper losses for the half-wave-inverter-based PMBDC machine compared to the full-wave-inverter-based PMBDC machine. This drawback could be interpreted as the result of moving the switch conduction and switching losses from the converter to the

TABLE 9.4 Comparison of the PMBDC machine variables based on the half-wave and the full-wave inverter

Ratio Between the Half-Wave-Inverter-Based and Full-Wave-Inverter-Based PMBDC Machine Variables	(i)	(ii)	(iii)
Maximum speed ratio	1	1.414	1
Electromagnetic torque ratio	1	0.707	1
Resistance ratio	4	2	2
Stator copper losses ratio	2	1	1
Size and cost ratio	1	1	>1

machine. It is easier to cool fractional-horsepower machines without significant additional resources; the thermal mass and surface area per unit output watt are higher in fhp machines compared to the integral-hp machines. Therefore, half-wave converter drives might be ideally suitable in fhp sizes. Further, the increase in cost to handle the thermal effects of higher copper losses in the machine has to be viewed from the overall perspective of the total cost of the motor drive system. The option (ii) is very preferable, to inherently exploit the full dc-link voltage in the half-wave-converter-based PMBDCM drive as compared to its counterpart and therefore to achieve a higher speed. That such a characteristic might be possible to exploit in many pump and fan drive applications is to be noted, thus making the half-wave-based PMBDCM drive an attractive technical solution. If the route through option (ii) is taken for comparison, a number of choices come into play, but they are not looked into for lack of space. Option (iii) might be the least desirable *prima facie* but has to be viewed in terms of the overall cost of the PMBDCM drive system to assess its suitability for a given application.

E. Impact of the Motor Inductance on the Dynamic Performance

From the machine equations, it is seen that the self-inductance of the phase winding plays a crucial role in the dynamics of the current loop and hence in the torque generation.

In the case of the full-wave-operated PMBDC machine, the electrical time constant is given by

$$\tau_{fw} = \frac{(L_s - M)}{R_s} \quad (9.241)$$

where M is the mutual inductance.

Consider a PMBDCM with twice the number of turns per phase as compared to the full-wave-operated machine, for operation with the half-wave converter and its self-inductance in terms of L_s is equal to $4L_s$ and its resistance for equal copper losses is $2R_s$, thus giving its electrical time constant as

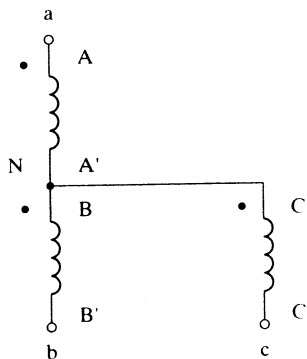
$$\tau_{hw} = \frac{4L_s}{2R_s} = \frac{2L_s}{R_s} \quad (9.242)$$

To find the ratio between these two time constants, it is necessary to express M in terms of L_s , from which we get

$$\frac{\tau_{hw}}{\tau_{fw}} = \frac{2L_s}{(L_s - M)} = \frac{2L_s}{L_s - \frac{1}{2}L_s} = \frac{2}{1 - \frac{1}{2}} \quad (9.243)$$

TABLE 9.5 Ratio of electrical time constants for various designs of the PMBDCM

Stator Slots per Pole per Phase	Ratio of Mutual to Self-Inductance, k_m	Ratio of Time Constants τ_{hw}/τ_{fw} for Turns per Phase		
		$2N_b$	$\sqrt{2} N_b$	N_b
1	-0.333	1.500	1.050	0.750
2	-0.400	1.428	1.010	0.714
3	-0.415	1.413	0.999	0.706

**Figure 9.57** New machine winding connections for the half-wave converter topology

where

$$M = k_m L_s \quad (9.244)$$

and k_m is used to evaluate the ratio of the time constants, given in Table 9.5 for a PMBDCM with turns per phase twice, $\sqrt{2}$, and equal to those of the full-wave-operated PMBDCM denoted by N_b . From the table, it is seen that the equivalent PMBDCM for operation with half-wave converter has the same electrical time constant compared to the full-wave PMBDCM drive, thus making the proposed drive suitable for very high-performance applications.

F. Winding Connections

The motor windings have to be connected as shown in Figure 9.57 for the split-supply converter as opposed to the half-wave converter connection, with all the phase wires forming the neutral having the same polarity for each winding. These connections will not create any additional manufacturing or design burden, thus having no impact on the cost.

G. Drive-System Description

The speed-controlled PMBDCM drive system schematic is shown in Figure 9.58. The feedback signals available for control are the phase currents, discrete rotor position signals from the Hall sensors to generate the gating instances for the phase switches, and rotor speed signal from a tachogenerator or from the position signal itself. The inner current loops enforce current commands, and the outer speed loop enforces the speed command. The speed signal is passed through a filter, and the resulting modified speed signal is compared with the speed reference to produce the speed-error signal. The

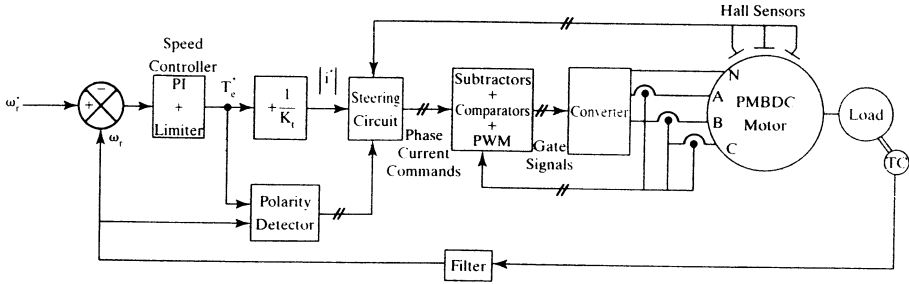


Figure 9.58 Schematic of the speed-controlled PMBDC motor drive system

torque-command signal is obtained from the speed-error signal through a speed controller that is a proportional-plus-integral (PI) type. The current magnitude reference is derived from the torque reference by a divider circuit, and the phase current commands are generated in combination with respective Hall position sensor signals through a steering circuit. The gating signals are generated for each phase switch by obtaining the phase current errors and processing these errors with PI current controllers and then combining them with carrier frequency to generate the pulse-width modulated signals.

H. Modeling, Simulation, and Analysis of the PMBDC Drive System

The various subsystems shown in Figure 9.58 are taken up for modeling, simulation, and analysis in this section.

Modeling of the PMBDCM with Converter Modes: PMBDCM model in abc phase variables is used in this simulation. Further, an ideal model with zero conduction voltage drop and zero switching times is utilized in this simulation for the switches and diodes. The operational modes determine whether one phase or two phases conduct at a given time; accordingly, the system equations emerge. To model when only phase A is conducting, the system equation is given by

$$R_s i_{as} + L_s \frac{di_{as}}{dt} + e_{as} = V_s \quad (9.245)$$

where R_s is stator phase resistance, L_s is self-inductance of a phase, e_{as} is a phase-induced emf, p is the derivative operator, and V_s is the dc-link voltage of the top half. The induced emf is given by

$$e_{as} = K_b f_{as}(\theta_r) \omega_r \quad (V) \quad (9.246)$$

where f_{as} is a unit function generator to correspond to the trapezoidal induced emf of the PMBDCM as a function of θ_r (which is the rotor electrical position). K_b is the emf constant, and ω_r is the rotor electrical speed. f_{as} is given by

$$\begin{aligned} f_{as}(\theta_r) &= (\theta_r) \frac{6}{\pi}, & 0 < \theta_r < \frac{\pi}{6} \\ &= 1, & \frac{\pi}{6} < \theta_r < \frac{5\pi}{6} \\ &= (\pi - \theta_r) \frac{6}{\pi}, & \frac{5\pi}{6} < \theta_r < \frac{7\pi}{6} \\ &= -1, & \frac{7\pi}{6} < \theta_r < \frac{11\pi}{6} \end{aligned} \quad (9.247)$$

For mode 3, when the phase a is being commutated and phase b is energized, the following equations apply.

$$R_s i_{as} + L_s p i_{as} + M p i_{bs} + e_{as} = -V_s \quad (9.248)$$

$$R_s i_{bs} + L_s p i_{bs} + M p i_{as} + e_{bs} = V_s \quad (9.249)$$

where the subscript b corresponds to respective phase b variables and parameters defined above. The f_{bs} is similar to f_{as} but phase-displaced by 120° electrical. Similarly, the equations for other modes can be derived from (9.245) to (9.249).

The electromechanical equation with the load is given by

$$J \frac{d\omega_m}{dt} + B\omega_m = T_e - T_l \quad (9.250)$$

where J is the moment of inertia, B is the friction coefficient, T_l is the load torque and T_e is the electromagnetic torque, given by

$$T_e = \lambda_p \{ f_{as}(\theta_r) i_{as} + f_{bs}(\theta_r) i_{bs} + f_{cs}(\theta_r) i_{cs} \} \quad (9.251)$$

The rotor position is derived for simulation from the following equation:

$$p\theta_r = \omega_r \quad (9.252)$$

Modeling of the Speed Controller: The speed controller is modeled as a PI type:

$$G_s(s) = K_{ps} + K_{is}/s \quad (9.253)$$

where s is Laplace operator, and from this equation the torque reference and current magnitude reference may be derived. This equation is written in time domain for simulation.

Steering Circuit: The steering circuit consists of three inputs, of which one is the current magnitude reference and the other two are polarity signals of the rotor speed and torque reference, denoted as $|i^*|$, ω_{rp} , T_{cp} , respectively. Depending on the polarity of the rotor speed and torque reference, the quadrant of operation and the phase sequence are determined as given in the Table 9.6.

Depending on the quadrant of operation and the position of the rotor, which determines induced emfs in the machine phases and hence the emf functions, the phase current commands are determined as follows.

Quadrant I	Quadrant IV	
$f_{as}(\theta_r) \geq 1, i_{as}^* = i^* $	$f_{as}(\theta_r) \leq -1, i_{as}^* = i^* $	
$f_{bs}(\theta_r) \geq 1, i_{bs}^* = i^* $	$f_{bs}(\theta_r) \leq -1, i_{bs}^* = i^* $	
$f_{cs}(\theta_r) \geq 1, i_{cs}^* = i^* $	$f_{cs}(\theta_r) \leq -1, i_{cs}^* = i^* $	(9.254)

Similarly for the third and fourth quadrants, the phase current commands are derived. This completes the modeling of the steering circuit.

Current-Loop Modeling: For phase A , the current-loop model including the pulse-width modulation is derived in the following, but that the same algorithm is

TABLE 9.6 Operational quadrant relationships

ω_{rp}	T_{ep}	Quadrant	Phase Seq.
≥ 0	≥ 0	I	abc
≥ 0	< 0	IV	abc
< 0	≥ 0	II	acb
< 0	< 0	III	acb

applicable to all other phases is to be noted. The current error, say for phase *A*, i_{aer} , which is the difference between the current reference and actual current, is amplified and processed through a proportional-plus-integral (PI) controller very similar to that of the speed controller. The output of the current controller is limited to its maximum value, given as i_{max} . The duty cycle of the switch for one period of the PWM cycle is then given by

$$d = \frac{i_{aer}}{i_{max}}, \quad i_{aer} > 0$$

$$= 0, \quad i_{aer} < 0 \quad (9.255)$$

The on-time of the switch, T_{on} , is given by

$$T_{on} = dT_c = \frac{d}{f_c} \quad (9.256)$$

where f_c is the carrier frequency of the PWM and T_c is the period of the PWM cycle.

The switch conduction time is updated only once in a PWM cycle, in order to avoid multiple switching in a PWM cycle. The off-time for the phase switches is obtained from the difference between T_c and T_{on} .

Simulation and Analysis: The parameters of the PMBDCM drive system for dynamic simulation are given next.

Poles = 4 $hp = 0.5$

$R_s = 0.7 \, \Omega/\text{ph}$ $L = 2.72 \, \text{mH}$ $M = 1.5 \, \text{mH}$

$K_b = 0.5128 \, \text{V}/(\text{rad}/\text{sec})(\text{Mech.})$ $K_t = 0.049 \, \text{N}\cdot\text{m}/\text{A}$

$J = 0.0002 \, \text{kg}\cdot\text{m}^2/\text{s}^2$ $B = 0.002 \, \text{N}\cdot\text{m}/\text{rad}/\text{sec}$.

$V_{s1} = V_{s2} = V_s = 160 \, \text{V}$; $f_c = 2 \, \text{kHz}$; Speed (base) = 4,000 rpm,

Current (base) = 17.35 A, V (base) = 40 V, Torque (base) = $0.89 \, \text{N}\cdot\text{m}$

Torque (max) = $2 * \text{Torque}(\text{base})$; $I_{max} = 2 * \text{Current}(\text{base})$

Speed Controller: Proportional gain $K_{ps} = 20$. Integral gain $K_{is} = 1$

Current Controllers: Proportional gain $K_{pi} = 50$, Integral gain $K_{ii} = 5$

A full four-quadrant operation is simulated by giving a two-directional step speed reference of rated value. The key responses of the actual speed, torque reference, actual torque, phase *a* current and its reference, phase *b* current, and rotor position are shown in Figure 9.59, together with the speed reference, all in normalized units. Although no

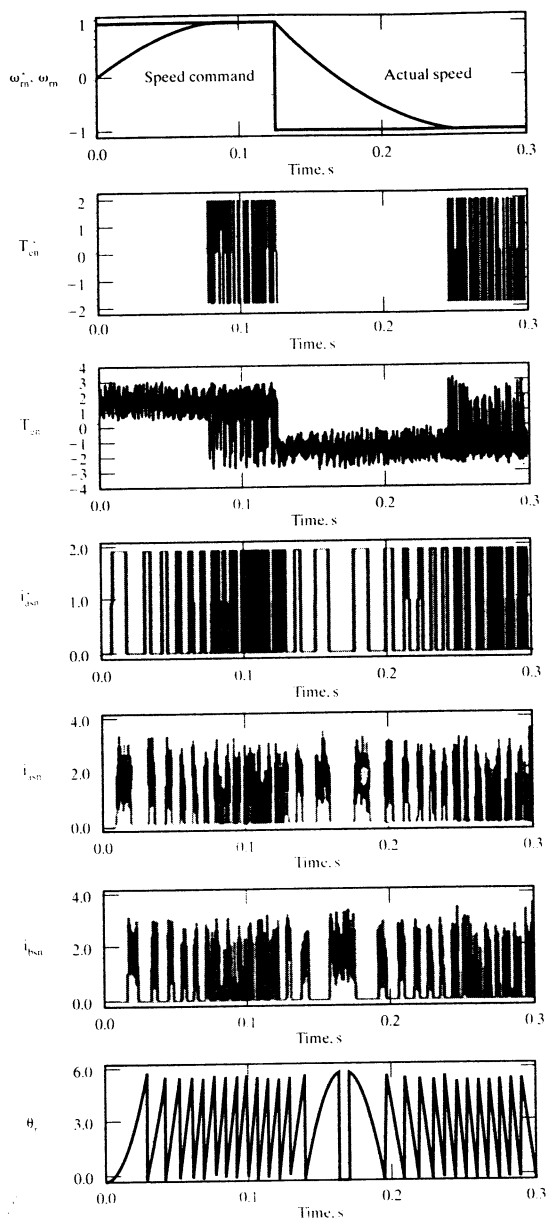


Figure 9.59 Dynamic simulation results of a four-quadrant PMBDC motor with the split-supply converter topology

TABLE 9.7 Comparison of the half-wave converter with the full-wave converter-based PMBDCM system

Aspects	Proposed Converter-based PMBDCM	Full-wave Converter-based PMBDCM
Number of power switches (all phases)	3	6
Number of diodes	3	6
Switch voltage (min)	$2V_s$	V_s
Switch current (peak)	I_p	I_p
Switch current (rms)	$I_p/\sqrt{3}$	$I_p/\sqrt{3}$
Motor ph. current (rms)	$I_p/\sqrt{3}$	$\sqrt{2}I_p/\sqrt{3}$
Number of capacitors	2	1
Capacitor voltage	V_p	V_p
Min. number of diodes in the front-end rectifier	2	4
Number of logic power supplies for isolated operation (min.)	2	4
Number of gate drivers	3	6
Switch conduction losses	$h v_{sw} I_p^*$	$2h v_{sw} I_p$
Diode conduction losses	$(1 - h) v_d I_p$	$2(1 - h) v_d I_p$
VA rating (peak)	$6V_s I_p$	$6V_s I_p$
VA rating (rms)	$2\sqrt{3}V_s I_p$	$2\sqrt{3}V_s I_p$

*Note: This current may be higher because of circulation energy.

steps have been taken to optimize the speed and current controllers, it becomes clear from the simulation results that the drive system is distinctly capable of four-quadrant operation. Further, it shows that the current response is fast enough for consideration of this drive in high-performance applications.

I. Comparison of the Half-Wave and Full-Wave Inverter-Based PMBDCM Drives

Consider the average duty cycle of the switch during a phase conduction as h , for deriving a comparison between the half-wave and the full-wave inverter-based PMBDCM drive system given in Table 9.7. Even though the VA ratings of the converters are the same, note that fewer devices with higher voltages usually cost less with the additional advantages of lower conduction and switching losses in the split-supply converter topology. The comparison does not consider the regenerative brake, which is required for both but with different VA ratings.

9.10.7.2 C-dump topology. The half-wave converter topology with one switch per phase has the disadvantage of utilizing only half of the available dc-link voltage. This fact precludes its use where high performance is a requirement. This could be resolved if topologies with more than one switch per phase but less than two switches per phase are resorted to. Such topologies have been developed for switched-reluctance motor drives with considerable success. One such topology is the C-dump converter with $n+1$ switches for an n -phase machine. The C-dump

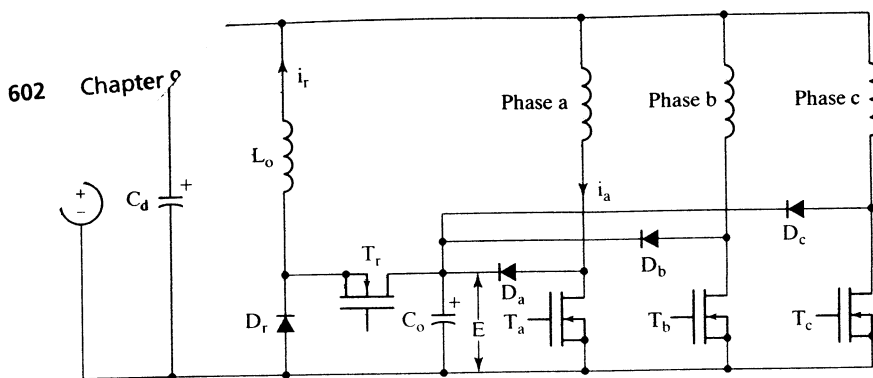


Figure 9.60 C-dump converter topology

topology is shown in Figure 9.60 for a three-phase machine. The principle of operation, analysis, and design of such a converter topology for a four-quadrant PMBDCM drive system are presented in this section. Design considerations for the PMBDCM to work with the C-dump power converter, based on power and torque equality with full-wave operated PMBDCM, are developed. A comparison of the C-dump and full-wave PMBDCM drives is derived to highlight key advantages and disadvantages of the C-dump-operated PMBDCM drive system.

A. Principle of Operation of the C-Dump PMBDCM Drive System

The C-dump converter for a three-phase system shown in Figure 9.60 is considered. It has four power switches and four power diodes with one of each for each phase winding and one set for energy recovery from the capacitor, C_o . Since the phase has only one switch, the current in it could only be unidirectional; hence, it is very similar to the half-wave converter-driven PMBDCM in operation. The motoring (I-quadrant) and regenerative (IV-quadrant) control of the C-dump-based PMBDCM are briefly described in the following.

Motoring Operation Assume that the direction of the motor is clockwise, which may be considered positive with the phase sequence abc of the motor phase windings for this discussion. The motoring operation is initiated when the phase voltage is in the flat region, i.e., at constant magnitude for a fixed speed and with the duration of 120° electrical. The phase a is energized when the phase current is commanded by turning on switch T_a ; the equivalent circuit is shown in Figure 9.61(i). When the current error is negative, switch T_a is turned off, and the current in the phase a winding is routed through the diode D_a to the energy recovery capacitor, C_o shown in Figure 9.61(ii). During this time, a negative voltage of magnitude $(E - V_{dc})$ is applied across the machine winding, thus reducing the current and bringing the current error to positive. The average air gap power and the input power are positive, giving a positive electromagnetic torque, thus indicating the operation is firmly in the first quadrant of the torque-vs.-speed region.

The motoring operation in the counterclockwise (reverse) direction of rotation of the motor is similar, except that the phase sequence will be *acb* in the energization of the motor phase windings. That corresponds to the III quadrant of the torque-speed characteristics.

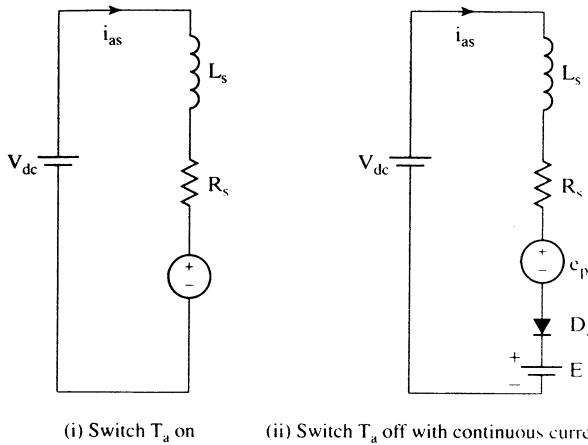


Figure 9.61 First-quadrant motoring operation with phase a of the PMBDCM drive

Regenerative Operation Whenever the energy has to be transferred from load to supply, the PMBDCM is to be operated as a generator, i.e., by generating a negative torque in the machine as against the positive torque for the motoring operation. It is usual to provide a current of opposite polarity to that of the induced emf in the full-wave converter-operated PMBDCM to generate a negative torque. It is not feasible in this C-dump-operated PMBDCM: it has unidirectional current feature for the positive half-cycle of the induced emfs. Then, the only alternative is to exploit the negative cycles of the induced emf, where only positive currents are required to obtain the negative torque. Such an operation for phase a involves the turn-on of T_a during negative constant emf period, and then, when the error current becomes negative, turning off T_a , enabling the conduction of D_a , resulting in the energy transfer from the machine phase a to the energy recovery capacitor. These operations are shown in Figure 9.62(i) and (ii). Note that the air gap power and the average input power to phase a are negative, indicating that the power has been transferred from the machine to the energy-recovery capacitor, C_o . The energy from C_o is recovered by a step-down chopper, using switch T_r and diode D_r shown in Figure 9.60. Note that this regenerative operation corresponds to the IV quadrant for a phase sequence of abc ; similar is the regenerative operation for reverse rotational direction of the PMBDCM, corresponding to the II quadrant.

B. Analysis of the C-Dump PMBDCM Drive

The analysis of the drive system with the C-dump topology is presented in this section. Effort is primarily made to obtain the maximum speed of the motor in terms of the duty cycle of the phase switches and the energy transferred to the energy-recovery capacitor, and hence an estimate of the power to be handled by the recovery chopper for a given motor rating. It is assumed that the commutation pulses are available through Hall sensors or encoders or resolvers or by estimation. Design guidelines and modeling can be found in reference [6].

Maximum Speed: Consider the machine voltage equation in steady state for rated stator current given by I_b ; it is given as follows:

$$V_{as} = R_s I_b + K_b \omega_r \quad (\text{V}) \quad (9.257)$$

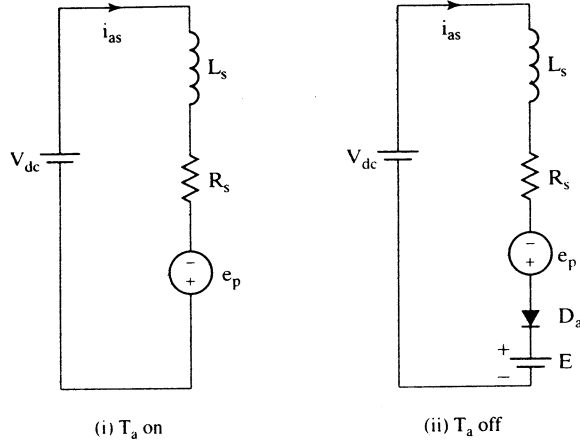


Figure 9.62 Operational mode and waveforms of variables for the IV quadrant regenerative operation of PMBDCM drive system

and the electrical rotor speed is obtained from this as

$$\omega_r = \frac{V_{as} - R_s I_b}{K_b} \text{ (rad/sec)} \quad (9.258)$$

For faster current-loop and hence torque and speed response, it is necessary to set aside some voltage, which is a fraction of the rated stator voltage, given as $k_a V_{as}$. Including this factor, the rotor speed is modified to

$$\omega_r = \frac{1}{K_b} [V_{as}(1 - k_a) - R_s I_b] \quad (9.259)$$

If an average duty cycle of the phase switches is denoted as h , then the stator phase voltage in terms of the dc-link voltage is written as

$$V_{as} = h V_{dc} \quad (9.260)$$

which, when combined with the rotor speed equation and normalization, yields the normalized rotor speed as

$$\omega_{rn} = h(1 - k_a) V_{dcn} - R_{sn} \text{ (p.u.)} \quad (9.261)$$

where the additional subscript n denotes the normalized values of the variables and parameters.

Typically k_a is in the range from 0.2 to 0.4, and h is varied from nearly zero to one. This relationship explicitly gives speed in terms of the duty cycle, dc-link voltage, stator resistance, and dynamic voltage reserve. This expression allows the determination of range of h variation for the desired variation of speed range. Determination of h is crucial to the evaluation of average energy-recovery current and hence in the rating of that circuit.

Peak-Recovery Current: The energy transferred to the energy-storage capacitor, C_o , during the turn-off intervals of phase switches has to be recovered through the energy-recovery circuit if losses are neglected. The average duty cycle of energy transfer from

the dc link and machine phase into capacitor C_o is $(1-h)$. Assuming that this stored energy is recovered through the chopper in a duty cycle of h , as this is essential to keep the separation of energy storage and recovery circuits, the powers can be equated as

$$E(1-h)I_p = EhI_r \quad (9.262)$$

where I_r is the peak recovery current through the chopper and could be written as

$$I_r = \frac{(1-h)}{h} \cdot I_p \quad (9.263)$$

As h increases, note that the energy recovered through the chopper reduces as I_r goes down, which in turn reduces the volt-ampere rating of the energy-recovery chopper circuit.

C. Comparison with Full-Wave Inverter-Controlled PMBDCM Drive

This section compares the C-dump-topology-based PMBDCM with that of the H-bridge inverter-fed PMBDCM drive system from the points of view of number of switches, passive components, their ratings, number of power supplies for gate drives, number of gate drives for isolated drive systems, converter losses, thermal management, and packaging requirements.

Some salient aspects and their comparison are given in Table 9.8. The average duty cycle for the phase switches is represented as h , and k_1 denotes the fraction of voltage for safety margin in the operation of the drive. Let k_2 be the fraction to give the regeneration brake current and

$$k_3 = \frac{E}{V_{dc}} \quad (9.264)$$

TABLE 9.8 Comparison of the C-dump and full-wave-based PMBDCM drives

Aspects	C-Dump-Based PMBDC	Full-Wave-Based PMBDC
Number of switch devices	4	7 (including regenerative brake)
Number of diodes	4	7
Switch voltage	E	V_{dc}
Switch peak current	I_p	I_p
RMS switch current	$\frac{I_p}{\sqrt{3}}$	$\frac{I_p}{\sqrt{3}}$
Motor phase rms current	$\frac{I_p}{\sqrt{3}}$	$\sqrt{\frac{2}{3}} I_p$
Power capacitors	2	1
Capacitor voltages	V_{dc} and E	V_{dc}
Number of logic power supplies (minimum)	2	4
Inductor	1	0
Number of gate drivers	4	7
Turn-off snubbers (if needed)	0	6
Switch losses	$v_{sw} I_p \left[h + \frac{1-h}{h} \right]$	$2h v_{sw} I_p$
Diode losses	$v_d I_p (1-h)$	$2v_d I_p (1-h)$
Total peak switch VA rating	$E I_p \left[3 + \frac{1-h}{h} \right]$	$6v_{dc}(1+k_1)I_p + v_{dc}(1+k_1)I_p k_2$

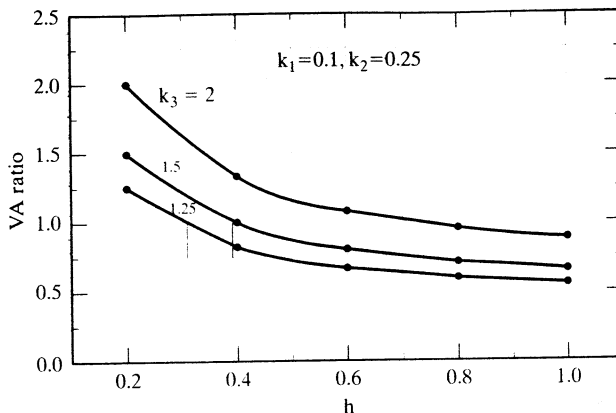


Figure 9.63 VA ratio vs. average duty cycle for various values of energy-recovery-capacitor voltages

This leads to the ratio of the VA rating as

$$\frac{VA_{cd}}{VA_{fw}} = \left[\frac{k_3}{1 + k_1} \right] \frac{\left(3 + \frac{1-h}{h} \right)}{(6 + k_2)} \quad (9.265)$$

For nominal values of $k_1 = 0.1$ and $k_2 = 0.25$, the relationship between the ratio of VA rating vs. the average duty cycle, h , for various values of k_3 ranging from 1.25 to 2 is shown in Figure 9.63. The break-even point for equal VA rating is shown with bold lines. Note that the duty cycle h is proportional to the normalized speed.

The converter switch losses are smaller for $h > 0.5$ in the case of the C-dump, and the diode losses are half those of the full-wave converter. This reduction in losses translates into heat sink and thermal management reduction by the same measure, resulting in a sizable reduction in packaging size. Further, it is helped by the requirement of smaller numbers of logic power supplies, snubbers, and gate drivers.

D. System Performance

The simulation takes the same drive-system parameters as given for the split-supply converter-operated drive system. The target value for the C-dump capacitor voltage is set to 175 V. The C-dump capacitor is initially charged to 100 V, the magnitude of the supply voltage. Figure 9.64 shows the commanded speed (ω_r^*), actual speed (ω_r), induced phase emf of phase a , phase- a current, air gap torque, and the voltage of the C-dump capacitor for the simulated speed loop with a speed command of ± 1000 r/min. The dump capacitor is charged to its target value of 175 V in less than 0.05 s. This voltage is then maintained in the proximity of 175 V. The transition from -1000 to $+1000$ r/min takes 0.9 seconds. As long as the commanded and actual speed are significantly different, the current command is at its maximum value of 20 A. Once the desired speed is achieved, the current drops to 8 A in order to support the motor load of 0.48 N·m. While the direction of rotation and the sign of the air gap torque are different, the load torque acts in unison with the air gap

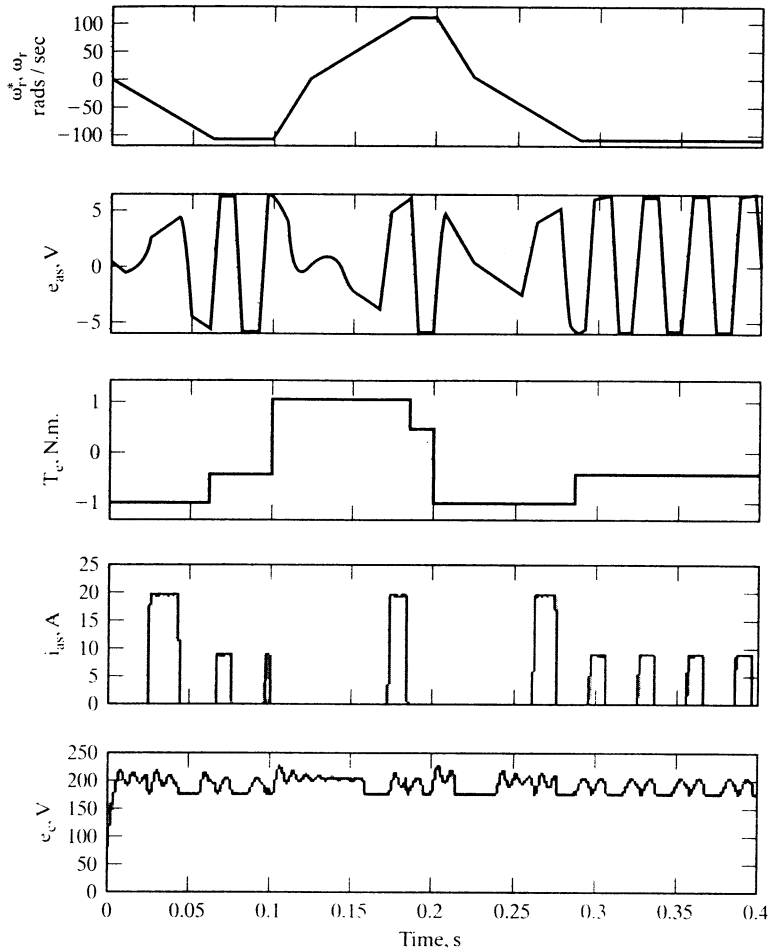


Figure 9.64 Dynamic simulation results of C-dump based PMBDCM drive systems

torque to decelerate the rotor. This results in faster deceleration of the rotor. The situation is different during acceleration, when the load opposes the air gap torque, resulting in slower acceleration than deceleration.

9.10.7.3 Variable-dc-link converter topology. One of the converter topologies with the advantage of varying the dc input voltage to the machine but with lower switch voltage not exceeding that of the dc source has other significant advantages also. Such a circuit is shown in Figure 9.65. In addition, this power converter topology has the advantages of the C-dump and split-supply converter topologies.

A. Principle of Operation

The converter circuit for a three-phase output has four switches and diodes, with the additional capacitor and inductor for its operation. The converter has two stages.

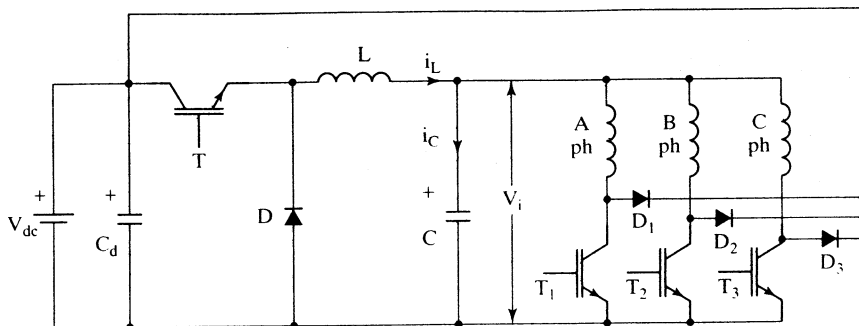


Figure 9.65 Variable-dc-link converter topology for PMBDC drives

The first stage is the chopper, which allows the variation of the input voltage to the machine. Switch T , diode D , inductor L , and capacitor C form a step-down-chopper power stage. The input voltage applied to the phases, V_i , is regulated by the operation of the chopper switch T . The inductor L and the capacitor C reduce the ripple content of the voltage V_i .

The second stage of the converter is the machine side, for handling the energy from the dc link to the machine and from the machine to the source. The chopper switch can be coordinated with the phase switches to regulate the current without having to switch the phase windings at carrier frequency. Because of the coordination option between the chopper switch and the phase switches, many modes of operation are possible in this drive system. The motoring (I quadrant) and regenerative (IV quadrant) control of the PMBDCM are briefly described with this converter.

Motoring Assume the direction of the motor is clockwise, which may be considered as positive with a phase sequence of abc of motor phase windings. The motoring operation is initiated when the phase voltage is constant positive for a fixed speed and with the duration of 120° electrical. Phase a is energized when the switch T_1 is turned on. To regulate current, T_1 is turned off, which initiates routing of the current through the freewheeling diode D_1 , source voltage V_{dc} , and capacitor C , applying a voltage of $(V_i - V_{dc})$ across the machine phases. The motoring operation is similar in the reverse direction, except that the phase energization sequence will be acb in the motor phase windings. This corresponds to quadrant III operation.

Regenerative Operation To transfer energy from the load to the source, the PMBDCM has to be operated as a generator, i.e., by providing negative torque to the machine. Negative torque is achieved by injecting a positive current during the negative constant-emf period.

On the basis of the rotor-position information and the polarity of i^* , the appropriate machine phase is turned on. The phase switch is turned on and modulated only if the current in the phase increases beyond a current window over the reference current, by hysteresis control. Normally, there is precise control of the machine input voltage through the chopper switch T , so the phase switch is rarely modulated to regulate the phase current. The phase switch is turned off only during commutation of the phase.

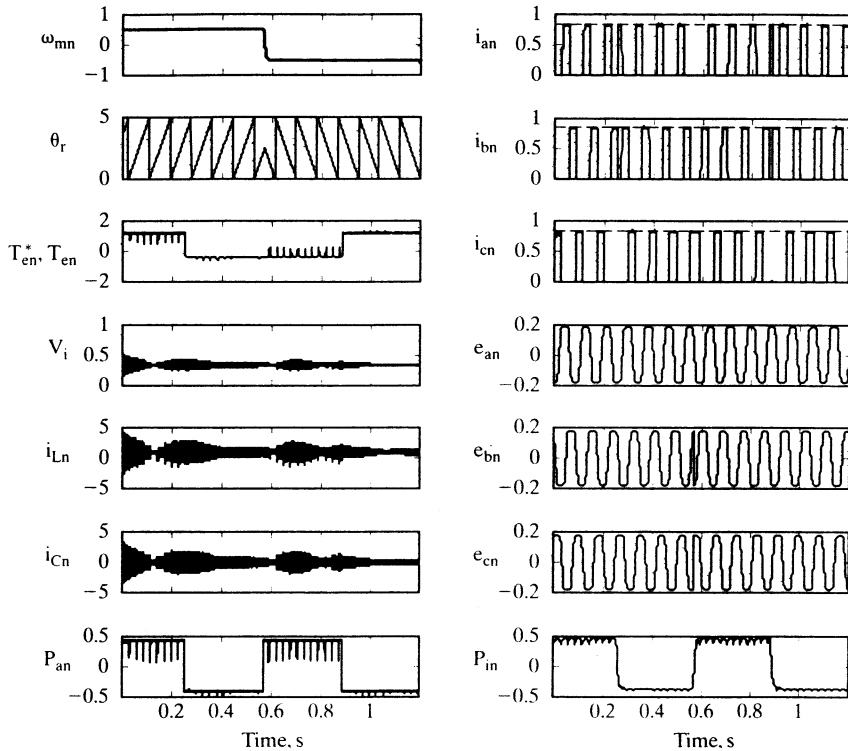


Figure 9.66 Torque-drive performance

B. System Performance

Consider the drive-system configuration as shown in Figure 9.58. For the torque-drive system, the speed-feedback loop is open. Both the torque- and speed-controlled drive systems are simulated, and results are given below.

Torque-Drive Performance Figure 9.66 shows the simulation results of torque-drive system performance, i.e., a system operating with only inner current loops. The machine is operating at 50% of the rated speed. The reference torque T_c^* is set at 1 p.u. by setting i^* to 1 p.u.. The phase currents i_{as} , i_{bs} , and i_{cs} are seen to reach the same value as the reference current i^* in the machine. It can be observed that the actual torque developed in the machine T_c has dips when there are transitions from one phase to another. Coordinating the currents in the incoming and outgoing phases can solve this problem. After 0.03 seconds, the commanded torque is switched from 1 p.u. to -1 p.u. while the speed remains the same, i.e., the system is operating in the IV quadrant. It can be observed that the voltage applied to the machine phases is about 0.5 p.u., unlike in other converter topologies, where a voltage switching between 0 and 1 p.u. is normally applied to the machine phases.

Speed-Controlled Drive Performance Figure 9.67 shows the simulation of a speed-controlled system operating in both forward and reverse directions. The current and torque are limited to 1.4 p.u. The current in the machine phases is unidirectional, but

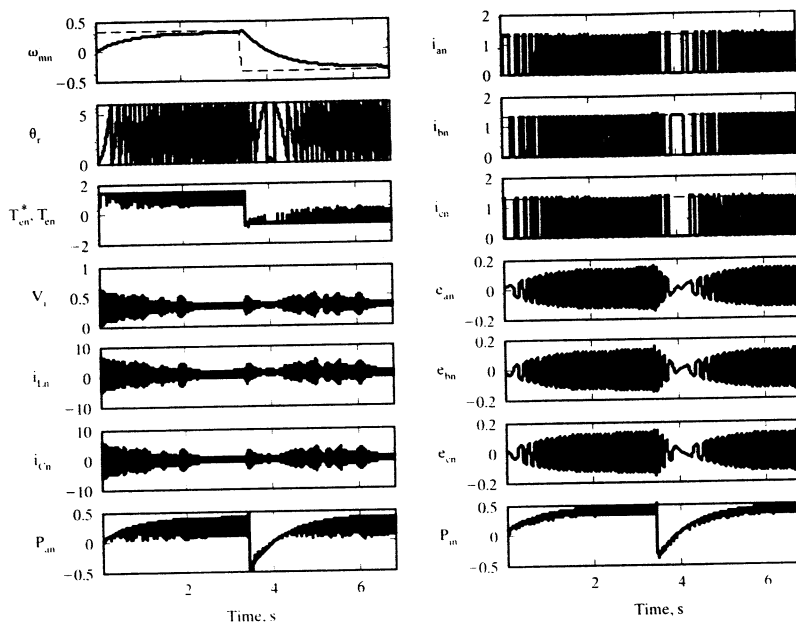


Figure 9.67 Speed-controlled-drive system performance

the torque developed by the machine is bidirectional. The input voltage to the phases, V_i , is varied with the speed. The current through the chopper inductor i_L has a high ripple content, but the currents in the machine phases have very little ripple. The variation in the ripple is due to machine inductances; additionally, it occurs because no subsequent switching occurs in the machine phases.

C. Merits and Demerits

The merits of the proposed topology are briefly listed next.

- (i) Only four switches and diodes are required for four-quadrant operation with a three-phase PMBDCM.
- (ii) Reducing gating driver circuits and logic power supplies saves money.
- (iii) Full four-quadrant operational capability is available.
- (iv) There is reduced possibility of shoot-through fault; switch is in series with the machine phase winding.
- (v) Capability of operating with one switch failure or one phase-winding failure raises reliability.
- (vi) Lower phase-switch losses raise efficiency.
- (vii) High-frequency ripple in steady-state operation can be considerably lower than that of the fixed-dc-link-based converters; this is a distinct advantage for high-performance drive systems. Unlike the C-dump converter, this con-

verter has no circulating energy, resulting in better efficiency and very low torque ripple.

- (viii) The power switches have a voltage rating equal to that of the source voltage, which is much lower than that of the C-dump and single-switch-per-phase converter.

The topology has the demerits associated with any half-wave converter topology, such as poorer utilization of the machine and a larger electrical time constant. In addition, this converter has two-stage power conversion, resulting in a slightly lower efficiency compared to single-stage power-converter topologies. Unlike the C-dump converter, this converter has no circulating energy, resulting in better efficiency and very low torque ripple.

9.10.8 Sensorless Control of PMBDCM Drive

The drive system is dependent on the position and current sensors for control. Elimination of both types of sensors is desirable in many applications, particularly in low-cost but high-volume applications, for cost and packaging considerations. Between the two sensors, the current sensor is easier to accommodate in the electronic part of the system; the position sensor requires a considerable labor and volume in the motor for its mounting. That makes it all the more important to do without the position sensor for the control of the PMBDCM drive system.

Current Sensing At least two phase currents are required for the current control of a three-phase machine. The phase currents can be sensed from the dc link current; hence, one sensor is sufficient for current control of the machine. The current sensors are relatively expensive if galvanic isolation is required. If isolation is not necessary, then the currents can be sensed inexpensively with precision resistors by measuring the voltage drops across them. The latter solution is used widely in low-cost motor drives. Another approach is to use MOSFET devices, with in-built current-sensing capability, to measure the currents. Alternatively, the MOSFET device itself serves as a sensing resistor during its conduction. The use of the drain-source voltage drop to estimate currents is fraught with inaccuracies due to temperature effect, and, for precise current control, the feedback from this voltage drop is not a viable method. Hall-effect current sensors are ideal for sensing the currents with galvanic isolation. At this stage, it is very nearly impossible to do away with current feedbacks for the control of the PMBDC machine to deliver high performance.

If precise torque and speed controls are not required, current feedback control and hence current sensing can be dispensed with. Then, a simple duty cycle using an open-loop PWM voltage controller is sufficient. However, the steering of the current to the appropriate machine phases requires the rotor-position information. A number of methods have come into practice to estimate rotor position without an externally mounted sensor.

Position Estimation Position can be sensed by Hall sensors overlooking a magnet wheel mounted on the shaft of the rotor extension with the magnets. This will provide just sufficient commutation signals, i.e., six per electrical cycle for a three-phase machine. Such a low discrete pulse count is not suitable for high-performance

applications. Optical encoders and resolvers provide the rotor position with high resolution, but they are expensive. Further, the position sensors require extensive mounting arrangements. High-volume applications demand that they be dispensed with, on account of the cost and manufacturing burdens. Many methods are possible to estimate the commutation signals; they are briefly described here.

- (i) Estimation by using machine model: The induced emf can be sensed from the machine model by using the applied currents and voltages and machine parameters of resistance, self-inductance, and mutual inductance. The advantage of this method is that an isolated signal can be extracted, because the input currents and voltages are themselves isolated signals. The voltages can be extracted from the base or gate drive signals and the dc-link voltage. The variations in the dc-link voltage can be estimated from the dc-link filter parameters and the dc-link current. Parameter sensitivity, particularly that of the stator resistance, will introduce an error in the induced emf estimation, resulting in inaccurate commutation signals to the inverter.
- (ii) Induced emf from sensing coils: Sensing coils in the machine can be installed inexpensively to obtain induced-emf signals. The advantages of this method are that the signals are fairly clean, parameter-insensitive, and galvanically isolated. The disadvantages are in the additional manufacturing process and additional wire harness from the machine. The latter is not acceptable in refrigerator compressor motor drives, because of hermetic sealing requirements.
- (iii) Sensing emfs from inactive phases: One of the most commonly used methods for acquiring position information is to monitor the induced emf of the machine phases when they are not being energized. Note that a machine phase is inactive for 33.33% of the time and that only two phases conduct at any given time. During the inactive time, an induced emf appears across the machine winding, which can be sensed. The induced emf of the phase yields the information on zero crossing and on when the emf reaches the constant region, indicating when that phase has to be energized. The polarity of the induced emf determines the appropriate polarity of the current to be injected into that machine phase. Instead of waiting for the constant region of the induced emf for energizing a machine phase, the induced emf on integration from its zero crossing will attain a particular value corresponding to thirty degrees from the zero crossing instant. The integrator output corresponding to thirty degrees from the positive zero crossing could be termed the *threshold value* used in energizing a phase. This threshold is independent of the rotor speed, as is shown below.

Assuming a trapezoidal induced emf whose peak is E_p at the rotor electrical speed of ω_b , the slope of the rising portion of the induced emf for any speed ω_s is given by dividing the peak value of the voltage at that speed by the time interval corresponding to thirty electrical degrees. Then, the instantaneous value of the induced emf during its rising instant is given by

$$e_{as}(t) = \frac{\left(\frac{E_p}{\omega_b}\right)\omega_s t}{(\pi/6\omega_s)} \quad (9.266)$$

which, upon integration from 0 to $\pi/6\omega_s$, yields the sensor output voltage, V_{vs} :

$$V_{vs} = \int_0^{\pi/6\omega_s} e_{as}(t) dt = \frac{\pi}{12} \frac{E_p}{\omega_b} \quad (9.267)$$

It can accordingly be proven that this algorithm also works for machines with sinusoidal induced emf, even though the sensor output voltage will be different. Note that the sensor output voltage is a constant, independent of the motor stator parameters, and that its magnitude is always the same for any operating speed of the machine. The only thing that could adversely affect the sensor output voltage is the induced-emf peak's decreasing with partial loss of rotor flux due to temperature sensitivity of the rotor magnets. This will clearly introduce errors, in that energization may not be exactly at thirty electrical degrees from the zero crossing as desired. Optimal utilization of the machine might not be possible in this condition, unless other corrective measures are taken.

- (iv) **Third-harmonic induced emf:** An alternative method is to detect the third-harmonic induced emf in the machine windings and use them to generate the control signals. A three-phase, star-connected, four-wire system will allow the collection of the third-harmonic induced emf, and this can be inexpensively instrumented with four resistors.

All the methods that rely on the induced emf have the disadvantage that, at standstill, the position information is not available, as there is no induced emf at zero speed. Even at very low speeds, the induced emf might not be easily detectable. Therefore, a method to generate the control signals at and around zero speed has to be incorporated for successful starting of the machine and up to a speed at which the induced-emf methods can come in to generate the position information reliably. Therefore, a starting procedure at standstill is required. This procedure can consist of two steps:

Step (i): Exciting one or two phases, the rotor can be aligned to a predetermined rotor position. This way, the starting position is known; hence, correct starting control signals are generated. When the rotor starts moving at slow speed, the induced emf is so small that it cannot be used for generating the commutation pulses until the rotor speed reaches a certain level. This fact necessitates a second step to complete the starting process.

Step (ii): Once the rotor starts moving, the stator phases are energized at a slowly varying frequency, keeping the stator currents constant. The rate of frequency variation is kept low so that synchronism is maintained and can be controlled modestly if the load is known *a priori*. If not, the stator frequency is altered by trial and error until it reaches the minimum speed at which the induced emfs are of sufficient magnitude to render them useful for control. This constitutes the second step in the starting process. The problem with this approach is that this is not precise; some jitter and vibrations can be felt during the starting, which may not be significantly adverse in many applications. In many cases, step (i) is skipped, and only step (ii) is used for starting of the machine.

A method based on the saliency of the rotor is another alternative, but caution must be used here: the saliency in the PMBDCMs is not very significant. This requires a detection of the machine inductance and its profile, from which the rotor position can be extracted.

9.10.9 Torque-Smoothing

It is not possible to generate ideal rectangular currents, because of the time delay introduced by the machine inductance. Therefore, the currents become more or less trapezoidal and produce a large commutation-torque ripple, as much as 10 to 15% of the rated torque. Further, the induced emfs are not exact trapezoids, because of significant slot harmonics. They, in turn, will generate harmonic ripple torques, resulting in poorer torque performance. The quality of the induced emfs is further affected by the type of armature winding. Windings are chosen for low-cost manufacturing for high-volume applications, and they invariably cause a greater deviation from the ideal waveforms. The cumulative effect of all these imperfections leads to a drive with uneven torque over an electrical cycle of its operation. That makes the drive highly unsuitable for high-performance applications. To overcome these disadvantages, methods based on current-shaping to counter the ill effects of the flux distribution are successful.

To overcome the unevenness in the flux-density distribution, it is measured or computed, and the current is continuously adjusted accordingly, to generate a constant torque. To counter the commutation-torque pulsation, the incoming-phase and outgoing-phase currents are coordinated in such a manner that the sum of the torque produced by the two phases is kept constant. All of the algorithms require a set of fast-acting current-control loops to shape the current, with no deviation either in magnitude or phase from their references.

9.10.10 Design of Current and Speed Controllers

Since the PMBDCM is similar to the separately-excited armature-controlled dc machine, as can be seen from its model, the methods that are appropriate to the design of the current and speed controllers for that can be gainfully employed here. The design of current and speed controllers directly relevant to this motor drive can be obtained from Chapters 3 and 4.

9.10.11 Parameter Sensitivity of the PMBDCM Drive

The motor parameters that are sensitive to variations in temperature are stator resistances and rotor magnets. The use of inner current loops overcomes the effect of stator-resistance variations. The use of speed-control loop counters the rotor flux-linkages variation. In that process, the linearity of the torque with its reference might be lost. In order to preserve the torque linearity in the drive system, methods similar to the air gap-power feedback control have to be resorted to.

The inductance variation is a function of saturation and hence of the exciting current. Therefore, it is easy to counter the inductance variations if the excitation current is measured and made available.

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9.12 DISCUSSION QUESTIONS

1. A PM synchronous motor can also be realized with permanent magnets on the stator and armature windings on the rotor. Discuss its merits and demerits compared to a PMSM with magnets on the rotor.
2. A cage rotor with PMs (called a *line-start PM synchronous motor*) can be used to start the motor as an induction motor and run it as a synchronous machine at utility frequency. Such an arrangement will improve the efficiency of the motor in comparison to induction motors. Discuss its construction, detailed operation, and possible applications.
3. The line-start PMSMs can also be operated from an inverter. In that case, the cage rotor will act as damper windings to damp out oscillations and, in case of loss of synchronization, the machine can run as an induction motor without disrupting the motion. Discuss an application scenario for such a motor drive. (Hint: High-reliability propulsion applications.)
4. Synchronous machines with surface-mount magnets have very little difference between direct-axis and quadrature-axis inductances. Explain why.
5. Interior PMSMs are preferred for high L_q/L_d ratios. Where would such a feature find application?
6. If the rotor is made salient with neither windings nor PMs, then the machine is a synchronous-reluctance motor. Its stator is that of the conventional PM synchronous motor. Due to its difference in quadrature- and direct-axis path reluctances, a torque is produced for an armature excitation. Since its field excitation is not from PMs, it must come from stator excitation. Since the inverter gets only active power from the dc link, where will the reactive power be generated for the excitation of the machine?
7. For equal power rating of the synchronous reluctance and PM synchronous motor, will the ratings of their inverters be equal? Explain.

8. Is L_q greater than L_d in the wound-rotor salient-pole synchronous machine?
9. What is the consequence of $L_q > L_d$ in control of the PMSM? (Hint: Consider maximum torque generated per unit input current.)
10. High-speed PMSMs have the rotor enclosed in a stainless sleeve to restrain the magnets against centrifugal forces. Ideally, will there be losses in the sleeve?
11. If the stator currents are sinusoids at fundamental frequency superposed with harmonics, will there be losses in the magnet sleeves? If there are losses, will they increase with rotor speed?
12. Injecting ac rectangular currents into PMSM is used in low-performance applications. Give reasons for such a control strategy and discuss the disadvantages of this control.
13. Torque pulsation is one of the measures for evaluating the suitability of a motor drive for an application. For a critical application requiring minimum torque pulsation, which is a suitable candidate between PMSM and PMBDCM drive?
14. A large number of control strategies have been developed and discussed for PMSM drives. Consider a four-quadrant application requiring speed variation to a maximum of base speed only. Which control strategy will be ideal, based on each one of the following considerations?
 - (i) Simplicity in implementation.
 - (ii) Maximum utilization of the inverter or minimum rating of the inverter.
 - (iii) Optimal output of the machine.
 - (iv) Optimal voltage utilization of the inverter and machine.
15. For implementation of any control strategy, a mapping from the torque and flux commands to the stator current and torque angle has to be performed. Is it a good choice to recommend on-line computation for this mapping? If so, justify it in the context of present computational capability available with processors.
16. A look-up-table implementation is considered for the control of a PMSM and PMBDCM drive system. Discuss the merits and demerits of the performance obtained with this implementation. Relate the bit resolution and accuracy in stator current and torque with the memory requirement for the implementation for each motor drive.
17. The inner current-control loops can use stator currents in stator or rotor reference frames. Discuss the merits and demerits of using each of the reference frame currents for current-feedback control.
18. An interior PMSM is completely demagnetized by injecting a negative d axis current. What is the magnitude of stator current to achieve demagnetization?
19. An interior PMSM is completely demagnetized by injecting a negative d axis current. Assume also a q axis stator current in the machine during this operation. Determine the electromagnetic torque generated in the machine.
20. Can magnet flux linkages be reduced by stator currents?
21. Can magnet flux linkages be increased by stator currents?
22. Air gap flux linkage is varied for flux-weakening. Will this affect the stator flux linkages?
23. Discuss the effects of losing control over d axis current in the high-speed operational region in a PMSM.
24. What are the salient differences between the maximum-torque-per-ampere strategy and the constant-mutual-flux-linkages strategy?
25. Including the effects of core losses in the formulation of control strategy is desirable. How can it be achieved?

26. Core losses are constant in one of the following schemes for constant-speed but variable-torque operation in a PMSM drive:
 - (i) maximum-torque-per-ampere control;
 - (ii) constant-flux-linkages control;
 - (iii) constant-torque-angle control.
 Identify the control strategy that gives constant core losses.
27. Propose a method of measurement for L_d and L_q of a PMSM, using the model developed in the text. [Hint: (i) Connect two phases together in a star-connected machine. (ii) Lock the rotor for each measurement.]
28. Almost all flux-weakening control schemes are dependent on machine parameters. Is there a control method to weaken the flux independent of machine parameters? If so, how can it be achieved?
29. Is flux-weakening possible in surface-mount-magnet machines?
30. Discuss a discrete-IC-chip-based implementation of a PMSM-drive constant-torque-angle control strategy.
31. Is it possible to implement other control strategies with discrete IC chips?
32. Why are processor-based implementations popular, and what are their advantages over discrete-IC-chip-based implementations?
33. Parameter sensitivity of PMSM has been discussed in this chapter. Can the state of the rotor magnet flux linkages be used to approximately predict stator temperature? If so, justify it with reasoning.
34. How critical is it to have parameter adaptation for a torque-controlled PMSM drive?
35. In the flux-weakening region, eventually the current loops saturate and six-step voltage operation will result. Discuss the effects of such an operation.
36. "Resorting to six-step operation in the flux-weakening region will give an enhanced torque-vs.-speed characteristic compared to the current-controlled operating region." Is this true? Develop a justification.
37. Sensorless operation is desirable. Enumerate the reasons and explain them.
38. Starting with precision from any rotor position without position transducers is difficult with many of the sensorless control algorithms. Explain the underlying reason for this statement.
39. The direct-axis self-inductance of the PMSM in stator reference frames can be modeled as $L_d^s = L_1 + L_2 \cos 2\theta_r$. Could this information be used to identify the rotor position, θ_r ? [Hint: Ref. [21].]
40. A voltage-source PWM inverter applies a fundamental and a number of higher-order harmonics into a PMSM. The dominant-harmonic current can be detected, from which the inductance can be calculated. This is achievable, because the harmonic inductive voltage drops are dominant compared to the resistive voltage drops. From inductances, the instantaneous rotor position can be extracted for control. The following steps are involved in the detection algorithm:

$$v_h = L \frac{di_h}{dt}$$

$$L = \begin{bmatrix} L_{11} + L_2 \cos 2\theta_r & L_2 \sin 2\theta_r \\ L_2 \sin 2\theta_r & L_{11} - L_2 \cos 2\theta_r \end{bmatrix}$$

where $\mathbf{v}_h = [v_{qsh} \ v_{dsh}]^T$ and $\mathbf{i}_h = [i_{qsh} \ i_{dsh}]^T$.

$\mathbf{v}_h$ is the stator qd axis harmonic-voltages vector, and $\mathbf{i}_h$ is the stator qd axis harmonic-current vector. By measuring $\mathbf{v}_h$ and $\mathbf{i}_h$ and by using the inductances of the machine, θ_r can be estimated. What is the difference between this method and the one given in discussion question 39? {Hint: Ref. [22].}

41. A sensorless-control algorithm injects a voltage signal at high frequency into the d axis of the PMSM. The current response is correlated with the rotor position by using machine-model-based current estimation. From the rotor position, rotor speed is derived. Compare this method with the method described in discussion questions 39 and 40. {Hint: Ref. [23].}
42. What is the effect of parameter sensitivity on the sensorless methods described in discussion questions 39, 40 and 41?
43. The mutual flux linkage increases with stator current. Will this saturate the stator core?
44. In PMBDCM, the induced emfs might not be exactly trapezoidal. What can cause their distortion?
45. "PMBDCMs have surface-mounted magnets." Is this true?
46. Considering fundamentals of voltages and currents only, the operation of PMBDCM and PMSM is similar. Is this true? If so, justify.
47. Commutation-torque ripple poses a serious problem (i) only at low speeds, (ii) only at high speeds, or (iii) at all speeds. Which is true?
48. Phase advancing in PMBDCM is equivalent to flux-weakening in PMSM to enable high-speed operation. What are the factors limiting the advance angle?
49. Torque-smoothing is feasible if the flux-density waveforms are known as a function of rotor position. If they are not known, how can torque-smoothing be accomplished?
50. In order to increase the high-speed operational region in PMBDCM, phase voltages are applied up until 150 degrees. How does this enable higher speed of operation?
51. Sensorless-control methods in PMBDCM are mainly induced-emf-based. Are the sensorless methods of PMSM applicable to PMBDCMs?
52. Rotor magnet sensitivity to temperature is significant in PMBDCMs. How can it be compensated for? Why does it need to be compensated for?
53. "Half-wave controlled PMBDCMs can deliver 4-quadrant operation." Is it true?
54. Why consider half-wave operation of PMBDCMs as against full-wave-controlled PMBDCMs?
55. "Half-wave operation of PMBDCMs transfers the power-device losses to the machine-armature losses." Is this statement, in general, true?
56. Half-wave operation of PMBDCMs underutilizes the machine. How can that be corrected?
57. In cost considerations of the PMBDCM drive, which one of the following alternatives is to be aimed for?
 - (i) Lowering motor cost only;
 - (ii) Lowering converter and controller cost only;
 - (iii) Lowering the total system cost, i.e., (i) and (ii) combined.
58. Some of the PMBDCM half-wave converters are in use with switched-reluctance motor drives. A certain industrial firm manufacturing both these motor drives can achieve cost efficiency by using the same half-wave converter for both. How can this statement be justified?

59. All the half-wave converters for PMBDCMs are shoot-through-failure-proof. Justify this statement.
60. For better utilization of the PMBDCM with half-wave operation, the windings will have a larger number of turns. If the fill factor of the slots is a constant for both the half-wave and full-wave PMBDCMs, enumerate the consequences of increasing the number of turns in the half-wave PMBDCM.

9.13 EXERCISE PROBLEMS

1. (i) For a PMSM, derive the normalized machine equations assuming

$$\lambda_b = \lambda_{af}, \quad L_b = L_d, \quad I_b = \frac{\lambda_b}{L_b}, \quad \rho = \frac{L_q}{L_d}$$

- (ii) What is the advantage of this normalization basis?
2. Prove that maximizing the mutual flux linkages with respect to stator current phasor results in

$$I_{sn} = \frac{-\cos \delta}{\cos^2 \delta + \rho^2 \sin^2 \delta}$$

and that electromagnetic torque is,

$$T_{en} = \frac{-\rho^2 \cos \delta \sin \delta (\cos^2 \delta + \rho \sin^2 \delta)}{(\cos^2 \delta + \rho^2 \sin^2 \delta)^2}$$

Use the normalized model derived in problem 1(i).

3. Will the strategy given in problem 2 yield a better performance than the maximum-torque-per-ampere strategy? Compare the strategies by using the machine parameters used throughout the text. [Hint: Compute δ from the torque equation, and then I_{sn} in the above.]
4. Using the model in problem 1, find the torque and mutual flux linkages developed by using the maximum-torque-per-unit-current and torque-for-constant-mutual-flux-linkages strategy with mutual flux linkages fixed at 1 p.u. The parameter of the machine: $P = 2$, and resistance can be neglected. What is the base speed in each case for these operating points? Assume base voltage is 1 p.u.
5. The parameters of a star-connected, 6-pole, 1.5-kW, 9.2-A, 1500-rpm, 9.55-N·m, 3-phase PMSM are as follows:

$$R_s = 0.513 \, \Omega, L_d = 4.74 \, \text{mH}, L_q = 9.51 \, \text{mH}, B_l = 9.36 \times 10^{-4} \, \text{N·m/(rad/sec)}, \\ J = 0.01 \, \text{kg·m}^2, \text{Emf constant} = 0.0669 \, \text{V/rpm}, \text{Inverter input voltage} = 285 \, \text{V}.$$

- (i) Determine the maximum speed of the PMSM drive system.
- (ii) Without exceeding the stator rated current and inverter input voltage, what is the maximum speed at which rated power is delivered?

The stator resistive drop can be neglected in the calculations for (ii), and the maximum stator-phase peak voltage obtained through the inverter is 55% of the dc-link voltage.

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